

MUSIC GENRE CLASSIFICATION FOR PREDICTIVE ANALYSIS USING DEEP LEARNING

*A Project Report submitted in partial fulfillment of the requirements for the
award of the degree of*

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE & ENGINEERING

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**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SHRI VISHNU ENGINEERING COLLEGE FOR WOMEN(A)**

(Approved by AICTE, accredited by NBA, Affiliated to JNTU Kakinada)

BHIMAVARAM – 534 202

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CERTIFICATE

*This is to certify that the project entitled "**Music Genre Classification for Predictive Analysis using Deep Learning**", is being submitted by **A.Dhatri Sai Subha, A.Sathvika, K.Mouly Prasanna, G.Baby Sravya** bearing the **Regd. No's. 21B01A0508, 21B01A0503, 22B05A0506, 21B01A0560** in partial fulfillment of the requirements for the award of the degree of "**Bachelor of Technology in Computer Science & Engineering**" is a record of bonafide work carried out by her under my guidance and supervision during the academic year **2024–2025** and it has been found worthy of acceptance according to the requirements of the university.*

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ABSTRACT

The Music Genre Classification for Predictive Analysis project focuses on leveraging deep learning techniques to classify music genres from audio files. The system converts raw audio data into Mel spectrograms, a time-frequency representation that captures essential musical features. These spectrograms are then processed using a Convolutional Neural Network (CNN), which extracts spatial and temporal patterns to identify the genre. By using an optimized neural network architecture, the model achieves high accuracy in distinguishing between different musical styles.

To ensure reliable classification, the audio files undergo preprocessing steps such as chunking, spectrogram conversion, and resizing for consistency. Each audio file is segmented into smaller overlapping chunks to maximize the number of training samples, helping the model learn better generalizations.

The deep learning model is trained using categorical cross-entropy loss and the Adam optimizer with a carefully tuned learning rate. Dropout layers are incorporated to prevent overfitting, and max pooling layers help in dimensionality reduction while retaining crucial features. The training process involves multiple epochs where the model refines its predictions, improving classification accuracy.

To make the system interactive and accessible, a Streamlit web application is developed, allowing users to upload an audio file and receive real-time predictions. The interface provides a smooth user experience with features like live prediction, loading indicators, and data visualizations. The model's real-world performance is tested by processing new audio files and analyzing classification accuracy.

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1. INTRODUCTION

Music genre classification has long been a fundamental aspect of organizing and discovering music. Traditionally, this process relied on human judgment, where genres were assigned based on artist reputation, album metadata, and listener perception. However, with the explosion of digital music and the emergence of numerous subgenres, conventional classification methods struggle to keep up with the dynamic nature of modern music. The increasing demand for accurate and scalable genre classification has necessitated the adoption of artificial intelligence (AI) and machine learning (ML) techniques to automate and enhance this process.

This report explores an advanced music genre classification system that utilizes deep learning, particularly Convolutional Neural Networks (CNNs), to analyze audio data and determine its genre with high precision. Unlike traditional methods that rely on metadata or manual labeling, this system extracts meaningful features directly from audio files, such as frequency patterns, pitch, timbre, and rhythmic structures. These features enable the model to learn complex relationships within musical compositions, improving classification accuracy and adaptability to various genres and subgenres.

A key innovation in this project is the integration of a real-time, user-friendly interface using Streamlit. This interface allows users to upload audio files and receive instant genre predictions based on the trained machine learning model. The real-time processing capability enhances accessibility and usability, making it applicable for various domains, including music streaming services, recommendation systems, and academic research. By leveraging AI, this system ensures that users can accurately and efficiently classify music without relying on static, predefined genre labels.

The methodology of this project involves multiple stages, including dataset preparation, feature extraction, model training, evaluation, and deployment. The dataset comprises diverse musical compositions spanning multiple genres, allowing the model to generalize well across different styles. Feature extraction is performed using spectrogram analysis, enabling the CNN to capture essential audio characteristics. The training process optimizes the model's ability to distinguish between genres, while evaluation metrics such as accuracy, precision, recall, and F1-score assess its performance. The final model is deployed using Streamlit, ensuring a seamless and interactive user experience.

By implementing this AI-driven approach, the project aims to improve the accuracy, scalability, and efficiency of music genre classification. Traditional genre classification methods often fail to recognize evolving music trends, leading to inaccurate or outdated categorizations. In contrast, this ML-based system continuously learns from new data, making it highly adaptable to contemporary music trends. This advancement benefits not only casual listeners but also industry professionals, content creators, and

researchers seeking a deeper understanding of music classification. Furthermore, by automating the classification process, this project contributes to the broader field of AI-driven music analysis, setting a foundation for future advancements in intelligent music discovery.

In conclusion, this report presents a cutting-edge approach to music genre classification by leveraging machine learning and deep learning techniques. The integration of CNNs for feature extraction and classification, combined with a real-time user interface, enhances the accuracy and efficiency of genre identification. By bridging the gap between technology and music discovery, this system paves the way for innovative applications in music streaming, recommendation engines, and audio analysis. The subsequent sections of this report will provide an in-depth examination of the system's development, performance evaluation, and potential future enhancements, demonstrating its significance in modern music classification and recommendation systems.

2. SYSTEM ANALYSIS

2.1 Existing system

Existing systems for Music Genre Classification typically rely on basic machine learning models such as K-Nearest Neighbors(KNN), Logistic Regression, SVM. These models are used for classifying genres by analyzing specific extracted features from audio files such as MFCCs, chroma features and spectral properties from the CSV file.

Features of Existing System:

- Simplicity and ease of implementation
- Effective for limited datasets with well-defined genre distinctions.
- Lower computational requirements compared to complex models.

Drawbacks of Existing System:

- Limited accuracy and performance, especially with complex genres.
- Ineffective in handling large and diverse music datasets.
- Lack of ability to capture the nuanced patterns within audio signals, leading to misclassification in genres with overlapping characteristics.

2.2 Proposed System

The proposed **Music Genre Classification System** utilizes deep learning techniques to automatically classify the genre of an audio file. The system is designed to process raw audio data, extract relevant features, and predict the genre with high accuracy. This approach eliminates the need for manual genre labeling, making music classification efficient, scalable, and user-friendly.

The system employs a Convolutional Neural Network (CNN) model trained on a diverse dataset of audio files spanning multiple genres. To enhance classification accuracy, the audio is first converted into Mel spectrograms, which effectively represent its frequency and amplitude variations over time. These spectrograms serve as inputs to the model, allowing it to learn the distinct characteristics of each genre.

A Streamlit-based web interface is integrated to provide an interactive experience where users can upload an MP3 or WAV file and receive an instant genre prediction along with a confidence score. The interface also enables users to listen to the uploaded audio, ensuring a seamless and engaging classification process.

To improve performance, the system incorporates techniques such as chunk-based audio

processing, feature normalization, and real-time inference. The trained model continuously adapts to new data, ensuring that it stays relevant to evolving music trends. Future enhancements may include expanding the genre classification range, optimizing the deep learning model, and incorporating additional music analysis features for richer insights.

This proposed system is designed to benefit music streaming services, content recommendation platforms, and digital music libraries, providing accurate, fast, and automated genre classification with minimal user effort.

Drawbacks of Proposed System:

- **Misclassifications:** Some music genres share similar characteristics making the model difficult to distinguish between them accurately.
- **Dependence on Training Data:** The accuracy of the model is highly dependent on the quality and diversity of the training dataset.
- **Sensitivity to Audio Quality and Background Noise:** The model's performance can degrade if the input audio contains background noise, distortions.

Objectives:

- **Enhanced Feature Extraction:** Utilized spectrograms to transform audio signals into visual representations for better analysis, improving genre classification by capturing detailed frequency patterns unique to different genres.
- **Pattern Recognition:** Used CNN to identify intricate tempo and melody patterns unique to each genre, leveraging deep learning filters to capture both low and high-level audio features for better classification.
- **Higher Accuracy:** Achieved an accuracy of 90% by improving model performance through optimizing CNN layers and fine-tuning hyperparameters.
- **User-Friendly Interface:** Developed a user-friendly interface for audio file uploads, enabling genre predictions for a smooth user experience.

2.3 Feasibility Study

A Feasibility Study is a comprehensive analysis conducted to determine the practicality and viability of a proposed system. It assesses various aspects such as technical, economic, legal, operational, and scheduling feasibility to ascertain if the project can be successfully completed within defined constraints. The study helps stakeholders make informed decisions about whether to proceed with the project or explore alternative options based on the identified risks, costs, benefits, and potential outcomes.

Technical Feasibility:

It depends on the availability of suitable hardware and software. Since the model relies on deep learning techniques such as Convolutional Neural Networks (CNNs) for feature extraction, it requires significant computational power for both training and real-time inference. The use of TensorFlow and Librosa for processing audio files ensures that the necessary tools are available, making the implementation technically feasible. However, performance optimizations may be needed for real-time applications, especially in resource-limited environments.

Economic Feasibility:

This considers the cost of development and deployment. Since the model is built using open-source frameworks like TensorFlow, Streamlit, and Librosa, the financial burden is minimal in terms of software. Deploying the model on cloud-based platforms can provide scalability while managing expenses efficiently. Given these considerations, the project remains economically viable, especially if integrated into existing music streaming services or recommendation systems.

Operational Feasibility:

It assesses whether the system meets user needs effectively. The model aims to provide a user-friendly interface for genre classification, allowing users to upload an audio file and receive instant predictions. With Streamlit as the front-end, the system is simple and intuitive.

3. SYSTEM REQUIREMENTS SPECIFICATION

3.1 Software Requirements:

Software requirements are crucial for ensuring the successful design, development, and deployment of the Music Genre Classification System. These requirements define the software tools and platforms necessary to build and execute the system, ensuring smooth functionality and performance. By specifying the software requirements, the development process becomes streamlined, allowing efficient collaboration among developers, testers, and end-users.

- Laptop/PC
- Python(3.10 or later)
- TensorFlow
- Keras (inside TensorFlow)
- Librosa
- Streamlit
- Visual studio code
- Browser
- Jupyter notebook

3.2 Hardware Requirements:

Hardware requirements are essential to ensure the efficient processing and training of the deep learning model. Since audio processing and model training consume high computational power, the hardware components must meet certain minimum specifications. This ensures seamless execution, faster processing, and a stable working environment.

- Processor (CPU): Intel Core i5 or higher
- Memory (RAM): 8 GB or higher
- Hard Disk (Storage): 256 GB SSD or higher
- Internet Connection: Required

3.3 Functional Requirements:

Functional requirements are the desired operations of a program, or system as defined in software development and systems engineering. The systems in systems engineering can be either software electronic hardware or combination software-driven electronics.

These are the requirements that the end user specifically demands as basic facilities that the system should offer. All these functionalities need to be necessarily incorporated into the system as a part of the contract.

- Audio File Upload
- Audio Splitting
- Feature Extraction
- Model Prediction
- Display Result
- Audio Playback
- Accuracy Indicator
- Home Page Display
- User Interface (UI)

3.4 Non-Functional Requirements:

Non-functional requirements define the quality attributes, performance, usability, and operational constraints of the system. These requirements ensure that the system performs efficiently under different conditions and provides a seamless user experience.

The following are the core non-functional requirements for this project:

- Scalability
- Accuracy
- User experience
- Modularity

4. SYSTEM DESIGN

Software design is the process of defining the methods, functions, objects, and overall structure of a software system to ensure that it meets user requirements. It involves planning the interaction between different components of the code to create a functional and efficient solution. A well-defined software design helps in achieving clarity, maintainability, and scalability in the development process.

System design, on the other hand, focuses on defining the architecture, components, modules, interfaces, and data flow within a system to fulfill specified requirements. It encompasses various aspects, including software, hardware, networks, and databases, making critical decisions regarding the structure, behavior, and performance of the system as a whole.

The importance of design and architecture in the early stages of a project cannot be overstated, as they establish the foundation for the entire development process. A well-planned design and architecture ensure alignment with project objectives, help in identifying potential risks, and provide a clear direction for implementation. Addressing these aspects early on allows teams to create a more structured and goal-oriented development roadmap.

Additionally, a strong system design aids in managing complexity, improving communication among team members, and enhancing collaboration. It serves as a blueprint for development, guiding implementation, testing, and maintenance. By investing in a comprehensive design approach from the outset, teams can create systems that are more scalable, maintainable, and capable of meeting long-term business needs.

Furthermore, defining system architecture at the beginning of a project enables a systematic and structured approach to development. It allows teams to anticipate and mitigate technical challenges, such as performance bottlenecks, scalability concerns, and integration issues, before they become major problems. By following established design principles and architectural patterns, teams can streamline development efforts, improve code quality, and reduce the need for extensive rework. Moreover, a well-structured architecture supports future enhancements and technological advancements, ensuring the system remains adaptable and sustainable over time. Ultimately, prioritizing design and architecture in the early phases of development leads to greater efficiency, lower risks, and a more successful software solution.

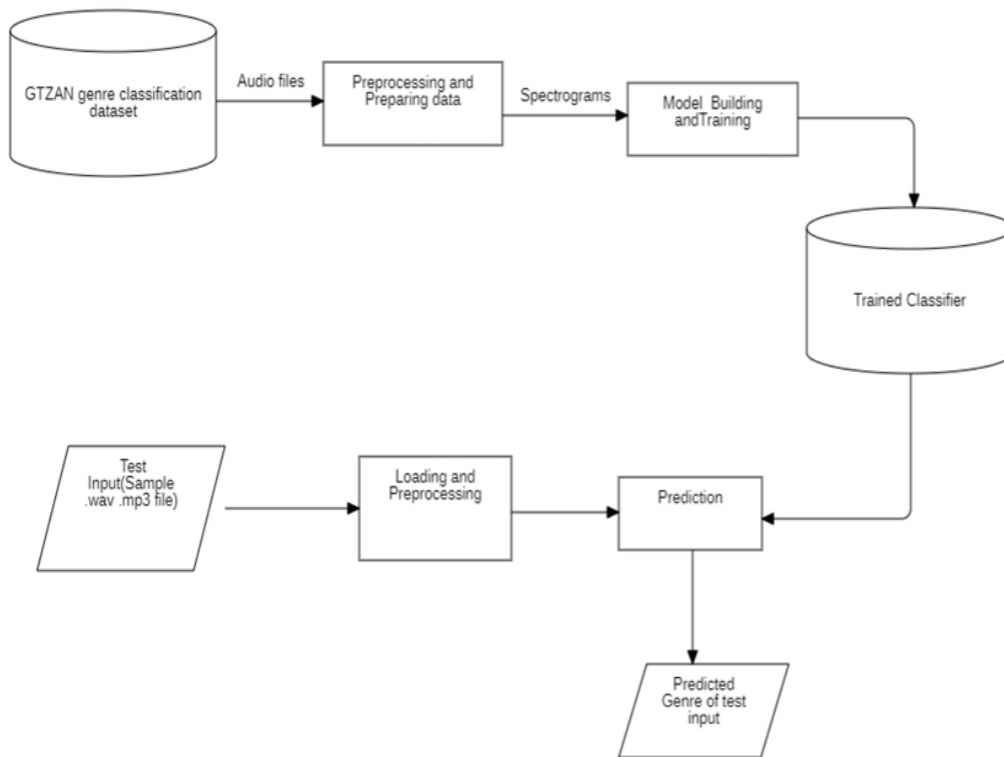


Fig 4.1: Flow diagram

n the above diagram,

- The process starts with loading the GTZAN genre classification dataset, which contains audio files.
- These audio files undergo preprocessing, where they are prepared and converted into spectrograms to extract frequency-based features.
- The spectrograms are used to train a machine learning model through model building and training.
- Once the model is trained, it is stored as a classifier for future predictions.
- For testing, a user uploads an audio file in .mp3 or .wav format.
- The uploaded file is loaded and preprocessed to generate its spectrogram representation.
- This processed data is passed through the trained classifier for prediction.
- The model predicts the genre based on the extracted features.
- The predicted genre of the test input is displayed to the user.
- The system ensures efficient data handling, accurate classification, and a user-friendly interface for seamless genre prediction.

4.1 Introduction to UML

Unified Modeling Language (UML) is a standardized diagrammatic language used in software development to visually represent system designs and structures. It provides a variety of diagrams and notations that help developers, analysts, and stakeholders understand different aspects of a system, such as its architecture, behavior, and interactions. Acting as a universal communication tool, UML ensures that all parties involved in software development can clearly interpret and discuss system requirements, improving collaboration and efficiency.

The use of UML in software development is essential for multiple reasons. Firstly, it helps in defining and formalizing system requirements through visual representations, reducing ambiguity and ensuring a shared understanding of functionality among stakeholders. Secondly, UML diagrams act as structured blueprints that guide the development process by clearly outlining how different system components interact and function. This structured approach aids developers in translating conceptual designs into actual code, resulting in more accurate and efficient software development.

Moreover, UML promotes modular design by allowing complex systems to be divided into smaller, manageable parts. This modular approach improves system maintainability and scalability, making it easier to modify or extend the software as needed. Additionally, UML supports agile and iterative development methodologies, enabling teams to rapidly prototype and refine system designs. This adaptability allows for quicker adjustments based on feedback and evolving project requirements, ensuring that the final software solution meets user expectations effectively.

The difference between planning with and without UML is significant in terms of clarity, precision, and project efficiency. Without UML, planning often relies on textual descriptions or informal sketches, which can lead to inconsistencies, misinterpretations, and communication gaps among team members. This lack of structured visualization makes it difficult to grasp complex system designs, increasing the chances of misunderstandings and delays during development.

In contrast, incorporating UML into planning provides a well-structured and standardized approach to software design. UML diagrams offer clear visualizations of system architecture, interactions, and requirements, helping teams align on objectives more effectively. This structured representation enhances communication among stakeholders, streamlines decision-making, and allows for early identification of potential risks and

dependencies. Ultimately, planning with UML ensures greater clarity, consistency, and efficiency, leading to the development of higher-quality, well-structured software solutions.

4.2 UML DIAGRAMS

A UML (Unified Modeling Language) diagram is a graphical representation used in software engineering to illustrate a system's structure, behavior, and interactions. These diagrams provide standardized symbols and notations that help developers, analysts, and stakeholders visualize and document different aspects of software systems. By using UML diagrams, teams can better understand complex system designs, facilitate communication, and ensure clarity throughout the software development lifecycle.

UML diagrams serve as a universal language for software design, enabling teams to create detailed representations of system components and their interactions. These diagrams are widely used for modeling static structures, dynamic behaviors, and system workflows. They are essential for designing, analyzing, and documenting software architectures, ensuring that all stakeholders have a shared understanding of the system's design and functionality.

Purpose and Goals of UML

The primary goals of UML design are as follows:

- To provide users with an expressive and standardized modeling language to develop meaningful models.
- To support extendibility and specialization mechanisms, allowing core concepts to be expanded.
- To remain independent of specific programming languages and development methodologies.
- To provide a formal framework for understanding software modeling concepts.
- To encourage the growth of object-oriented (OO) tools and frameworks.
- To support high-level development concepts, such as design patterns, components, and frameworks.
- To integrate best practices from the software industry into a unified modeling approach.

Types of UML Diagrams

UML diagrams are categorized into three primary groups: structural diagrams, behavioral diagrams, and interaction diagrams. Each serves a unique purpose in system

design and development.

1. Structural Diagrams

Structural diagrams define the static aspects of a system, focusing on components, objects,

and their relationships. These diagrams provide a blueprint for understanding a system's architecture and organization.

Class Diagram

- Represents the static structure of a system by illustrating classes, attributes, methods, and relationships.
- Helps in understanding object-oriented design and serves as a foundation for system implementation.

Object Diagram

- Displays instances of classes and their relationships at a specific point in time.
- Useful for verifying class relationships and understanding instance-level interactions during runtime.

Component Diagram

- Depicts the physical components of a system and their dependencies.
- Focuses on software modules, libraries, and executables within the system.

Composite Structure Diagram

- Illustrates the internal structure of a class or component, showing how its parts are interconnected.
- Helps in modeling complex components or subsystems.

Package Diagram

- Represents the organization and dependencies of software packages within a system.
- Provides a hierarchical view of system modules, facilitating better organization and reuse.

Deployment Diagram

- Shows the physical deployment of software components on hardware nodes.
- Helps in understanding the distribution of system elements in a networked environment.

2. Behavioral Diagrams

Behavioral diagrams model the dynamic aspects of a system, focusing on processes, workflows, and state changes. These diagrams help visualize how the system functions over time.

Use Case Diagram

- Represents functional requirements by illustrating interactions between users (actors) and the system.
- Helps stakeholders understand different use cases and expected system behavior.

Activity Diagram

- Depicts the flow of activities and decision points within a system.
- Used to model business processes and system workflows.

State Machine Diagram

- Models the dynamic behavior of a system by showing states, transitions, and events.
- Useful for designing state-dependent logic within an application.

3. Interaction Diagrams

Interaction diagrams illustrate how different components interact over time to execute specific functionalities. These diagrams provide insights into message flows, collaborations, and process execution.

Sequence Diagram

- Shows the sequence of messages exchanged between objects over time.
- Helps in analyzing the flow of control within a system and identifying dependencies.

Communication Diagram

- Formerly known as a collaboration diagram, it focuses on object relationships and interactions.
- Provides a structural view of how objects communicate to fulfill a specific functionality.

Significance of UML Diagrams

UML diagrams play a crucial role in software development by providing a clear visual representation of complex systems. They facilitate communication among team members, ensure consistency in design, and serve as a reference throughout the development process. The benefits of using UML diagrams include:

- Improved Communication: UML diagrams act as a common language for developers, designers, and business stakeholders, reducing ambiguity in system design.

- Enhanced Documentation: These diagrams serve as detailed documentation, making it easier to maintain and scale software systems.
- Better System Analysis: By visually mapping out system components and interactions, teams can identify potential design flaws early in the development process.
- Efficient Development Workflow: With well-structured UML diagrams, development teams can streamline implementation, testing, and maintenance efforts.

Comparison: Planning with UML vs. Without UML

Planning software development without UML often leads to vague, inconsistent, or misinterpreted system designs. When teams rely solely on textual descriptions or informal sketches, they may encounter communication gaps, delays, and design flaws. Without a clear visual representation, complex system relationships and behaviors become difficult to understand, leading to errors and inefficiencies.

On the other hand, using UML diagrams in planning provides a structured, visual approach to software design. With well-defined diagrams, teams can ensure better clarity, consistency, and accuracy in their development process. UML diagrams enable teams to:

- Clearly define system requirements and interactions.
- Identify dependencies and potential risks early.
- Maintain a systematic and organized development approach.
- Foster better collaboration among team members and stakeholders.

Extending UML for Modern Software Development

As software systems evolve, UML has also adapted to modern development practices. UML can be extended and customized to suit various software methodologies, including Agile, DevOps, and Microservices Architecture. Some modern extensions of UML include:

- SysML (Systems Modeling Language): A UML-based language tailored for systems engineering.
- UML Profiles: Custom extensions that adapt UML for specific industries, such as healthcare, finance, or embedded systems.
- Integration with Code Generation Tools: UML can be used alongside tools that generate executable code, enhancing development efficiency.

UML diagrams are essential tools in software engineering, providing a visual representation of system structure, behavior, and interactions. By leveraging UML diagrams, development teams can enhance communication, improve design clarity, and

ensure an efficient software development process.

Whether for analyzing system requirements, designing robust architectures, or optimizing workflows, UML diagrams help in achieving higher-quality and well-structured software solutions. As an industry-standard modeling language, UML continues to be a valuable asset for software developers, engineers, and business analysts in creating scalable and maintainable software systems.

USE CASE DIAGRAM:

A Use Case Diagram is a type of behavioral diagram in UML that visually represents the functional requirements of a system from the perspective of its users or external actors. It provides a high-level overview of the system's functionalities and interactions between users and the system. These diagrams are widely used during the requirements analysis phase of software development to capture and communicate the system's intended behavior.

Components of a Use Case Diagram

Actors: Represent users, external systems, or entities interacting with the system, depicted as stick figures outside the system boundary.

Use Cases: Represent specific functionalities provided to users, depicted as ovals within the system boundary, and connected to actors interacting with them.

Relationships: Indicate interactions between actors and use cases, including:

Association: Represents communication between an actor and a use case.

Generalization: Indicates that one actor or use case inherits behavior from another.

Benefits of Use Case Diagrams

Requirements Analysis: Helps stakeholders document system requirements from a user's perspective.

Communication: Serves as a tool for developers, analysts, and stakeholders to discuss system functionality.

System Design: Aids in designing system architecture and user interface.

Testing: Provides a basis for defining test cases and validating system behavior.

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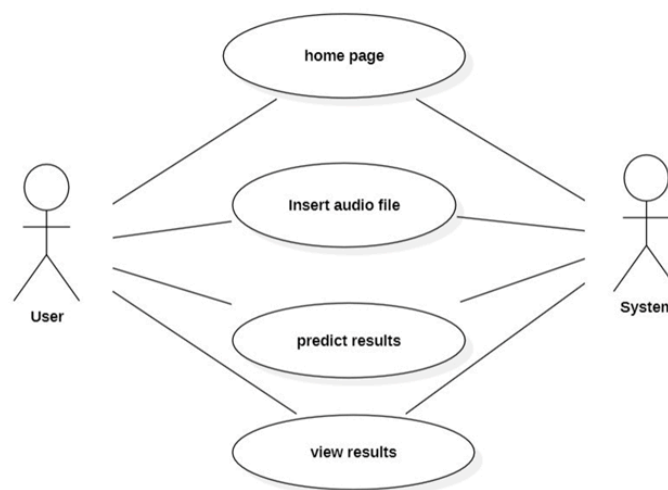


Fig 4.2.1: Usecase diagram

Sequence Diagram:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a message sequence chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

It represents the flow of messages in the system and is also termed as an event diagram. It helps in envisioning several dynamic scenarios.

It portrays the communication between any two lifelines as a time-ordered sequence of events, such that these lifelines took part at the run time.

In UML, the lifeline is represented by a vertical bar, whereas the message flow is represented by a vertical dotted line that extends across the bottom of the page.

It incorporates the iterations as well as branching.

Purpose of Sequence Diagram

- Model high-level interaction between active objects in a system
- Model the interaction between object instances within a collaboration that realizes a use case
- Model the interaction between objects within a collaboration that realizes an operation.

Sequence Diagrams at a Glance:

Sequence Diagrams show elements as they interact over time and they are organized according to object (horizontally) and time (vertically):

Object Dimension

- The horizontal axis shows the elements that are involved in the interaction
- Conventionally, the objects involved in the operation are listed from left to right according to when they take part in the message sequence. However, the elements on the horizontal axis may appear in any order.

Time Dimension

- The vertical axis represents time proceedings (or progressing) down the page. Note that:

Time in a sequence diagram is all about ordering, not duration. The vertical space in an interaction diagram is not relevant for the duration of the interaction.

Sequence diagrams are essential tools in software engineering, providing a clear and

structured representation of object interactions over time. They help developers, designers, and stakeholders understand how different system components communicate, making it easier to design, debug, and optimize workflows. By illustrating the flow of messages in a system, sequence diagrams enhance clarity, reduce ambiguities, and improve collaboration during the software development lifecycle. Their ability to depict real-time interactions makes them invaluable for analyzing system behavior, refining system architecture, and ensuring smooth execution of business processes.

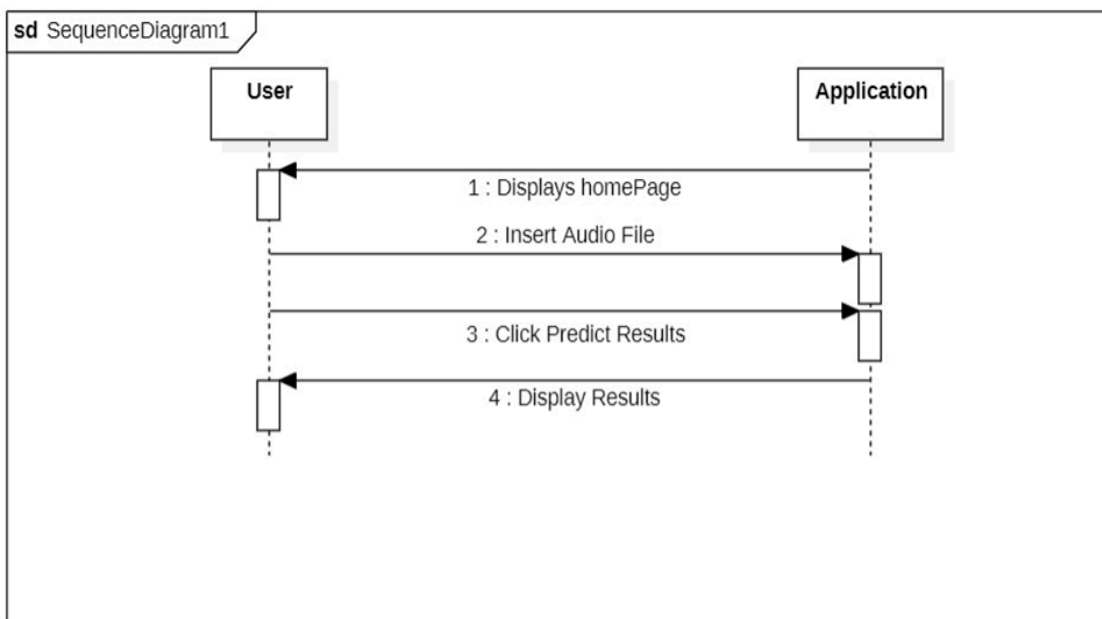


Fig 4.2.2: Sequence diagram

Activity diagram:

An Activity Diagram in UML (Unified Modeling Language) is a type of behavioral diagram that represents the workflow of a system, process, or business operation. It visually describes the sequence of activities performed in a system, including decision points, parallel operations, and conditions. Activity diagrams help in understanding how a system operates and are widely used in business process modeling, system analysis, and software design.

Purpose of Activity Diagrams

- Workflow Representation: They illustrate the sequence of activities within a system

or process.

- Process Modeling: Useful for modeling business workflows, system operations, and complex algorithms.
- Decision Making: Shows different paths based on conditions and decisions within a process.
- Parallel Processing: Helps represent concurrent processes occurring simultaneously.
- System Understanding: Aids in analyzing and improving system functionality and efficiency.

Components of an Activity Diagram

1. Initial Node: Represents the starting point of the activity.
2. Activity (Action) Nodes: Denotes specific tasks or operations performed within the process.
3. Decision Node: Represents a branching point where different paths can be taken based on conditions.
4. Merge Node: Combines multiple paths back into a single flow.
5. Fork Node: Splits a flow into multiple parallel activities.
6. Join Node: Merges parallel activities back into a single flow.
7. Final Node: Represents the end of the activity.
8. Transitions (Arrows): Show the flow of control between different nodes.

Advantages of Activity Diagrams

- Provides a clear and structured visualization of complex processes.
- Helps in identifying bottlenecks and inefficiencies in a system.
- Aids in designing automated workflows and business logic.
- Facilitates better communication between developers, business analysts, and stakeholders.
- Useful in testing and validation of system workflows.

Activity diagrams play a crucial role in understanding, designing, and optimizing

processes within software systems. They offer a visual representation of workflows, making it easier to analyze complex operations and improve efficiency. By clearly mapping out activities, decisions, and conditions, activity diagrams help teams collaborate effectively, ensuring a smooth and well-structured development process.

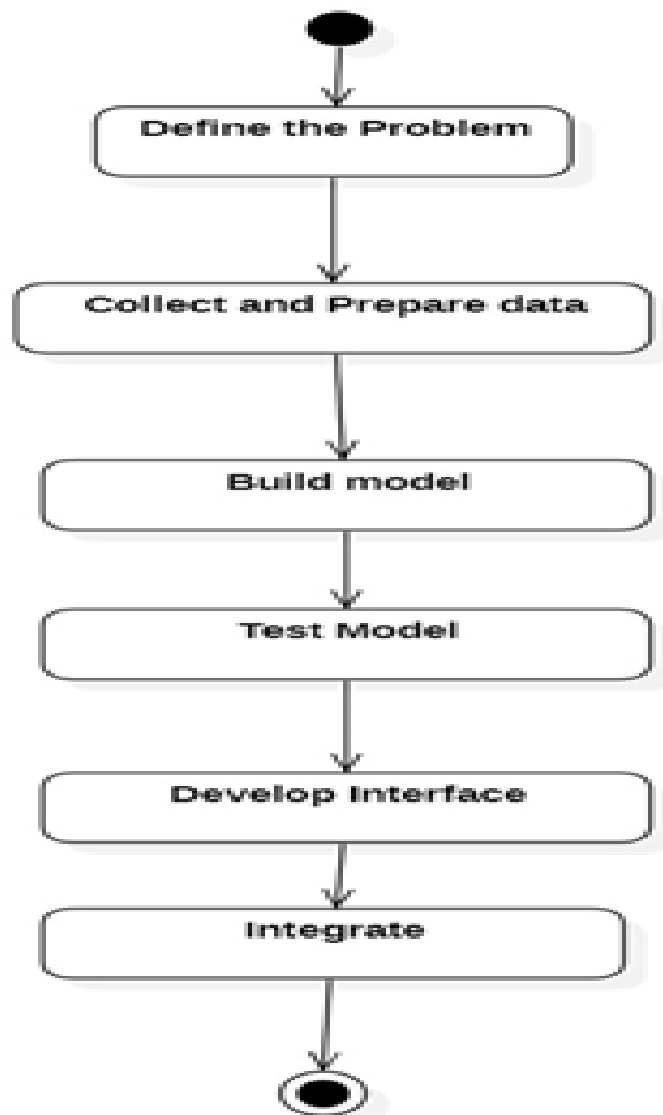


Fig 4.2.3: Activity diagram

CLASS DIAGRAM:

A Class Diagram is one of the most fundamental types of UML (Unified Modeling Language) diagrams used in software design. It provides a static view of a system by representing its classes, attributes, methods, and relationships among objects. Class diagrams are widely used in object-oriented programming (OOP) to visually model the structure of software systems.

Purpose of Class Diagrams

Class diagrams serve as a blueprint for designing and implementing object-oriented systems. Their main purposes include:

System Representation: They depict the static structure of a system, including classes, their properties, and associations.

Code Generation: Many programming languages use class diagrams as a reference for implementation.

Database Design: Helps in designing relational databases by mapping classes to tables.

Improved Understanding: Provides a clear and organized representation of system components for developers, designers, and stakeholders.

Facilitates Reusability: Encourages modular and reusable designs by clearly defining class responsibilities and interactions.

Components of a Class Diagram

A class represents a blueprint for objects. It contains attributes (data members) and methods (functions).

Classes are represented as rectangular boxes with three compartments:

Top section: Contains the class name.

Middle section: Lists attributes (variables).

Bottom section: Lists methods (functions).

Attributes

Represent the data or properties associated with a class.

Each attribute has a visibility modifier:

Public (+): Accessible from anywhere.

Private (-): Accessible only within the class.

Protected (#): Accessible within the class and its subclasses.

Methods (Operations)

Define the behavior of a class by performing specific functions.

Methods also follow visibility rules similar to attributes.

Relationships

Define the connections between classes. Common relationships include:

Association: A simple connection between two classes where one class uses another. Represented by a solid line.

Aggregation: Represents a "whole-part" relationship where one class is made up of others. Represented by a hollow diamond.

Composition: A stronger form of aggregation where the parts cannot exist independently. Represented by a filled diamond.

Inheritance (Generalization): Represents an "is-a" relationship where a child class inherits from a parent class. Represented by a solid line with a hollow triangle.

Dependency: A weaker relationship where a class depends on another but does not own it. Represented by a dashed arrow.

Advantages of Class Diagrams

Clear System Design: Provides a structured way to represent system components.

Efficient Communication: Helps developers, architects, and stakeholders understand the design easily.

Code Structure Visualization: Maps out the relationships between objects before implementation.

Simplifies Maintenance: Acts as documentation for future enhancements.

Encourages Reusability: Promotes modular programming and reusable components.

Class diagrams are essential for modeling the static structure of object-oriented systems. They help in designing software by clearly defining class properties, behaviors, and relationships. By using class diagrams, teams can ensure a well-organized system architecture that is easy to develop, maintain, and scale. Whether used for software engineering, database design, or business modeling, class diagrams provide a solid foundation for system development.

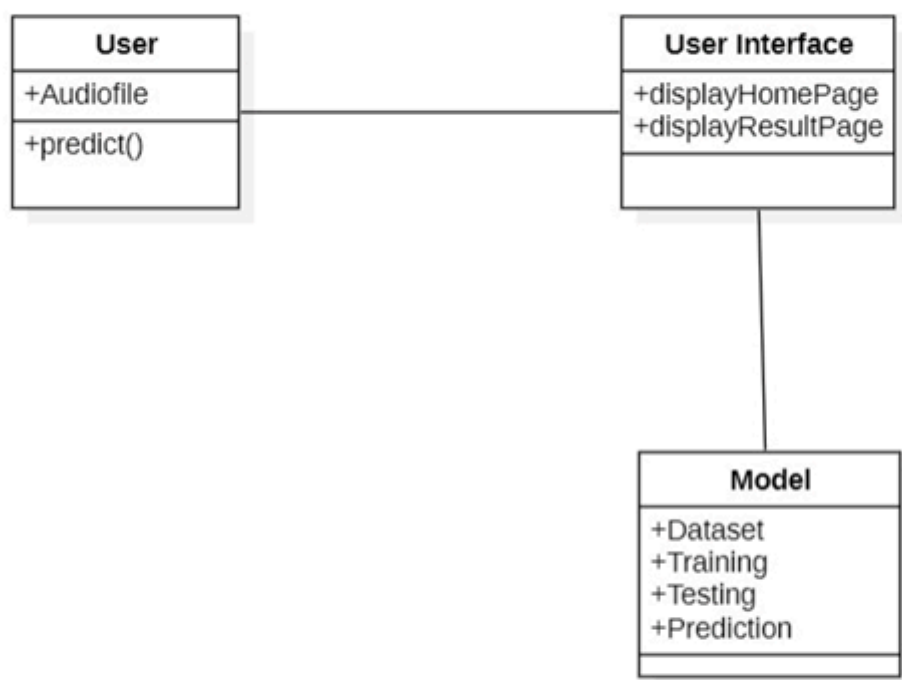


Fig 4.2.4: Class diagram

5. SYSTEM IMPLEMENTATION

System implementation refers to the process of integrating and deploying a designed system into a working environment, ensuring that it functions as intended. It involves converting the conceptual design into a fully operational system by following structured steps such as coding, testing, deployment, and maintenance.

During implementation, various components of the system, such as software, hardware, databases, user interfaces, and machine learning models, are configured and tested to ensure proper functionality. The process also includes data migration, security measures, and user training to enhance usability and performance.

System implementation can be executed using different methodologies, including incremental, phased, direct, or parallel implementation, depending on the complexity and criticality of the project. A well-planned implementation ensures minimal risks, reduces errors, and enhances system efficiency, leading to a successful deployment.

5.1 Introduction

The implementation of the Music Genre Classification System involves developing a machine learning-based model that can accurately classify the genre of a given audio file. The system is designed to process audio files, extract meaningful features, and use a trained deep learning model to predict the most probable genre. The entire implementation is structured to ensure accuracy, efficiency, and a seamless user experience.

The implementation process begins with data preprocessing, where the audio file is converted into a Mel spectrogram representation using Librosa, ensuring that relevant sound patterns are captured. These spectrograms are then resized to match the input dimensions of the model. The pre-trained Convolutional Neural Network (CNN) model is loaded, which performs genre classification based on learned patterns from the training dataset.

The system is integrated into a user-friendly interface using Streamlit, allowing users to upload an audio file and receive real-time predictions. The interface also provides additional functionalities, such as playing the uploaded audio and displaying the confidence score of the prediction. To enhance user engagement, animations like balloons are used upon successful classification.

The system is also designed to handle various audio formats while maintaining high classification accuracy. Future enhancements may include real-time classification, multilingual support, and advanced deep learning architectures to improve robustness and adaptability to evolving music trends.

Thus, the system implementation ensures a smooth integration of machine learning,

audio processing, and web-based interaction, making it a reliable and scalable solution for automated music genre classification.

5.2 Modules

Data Collection

The dataset for music genre classification consists of audio files categorized into ten different genres: blues, classical, country, disco, hip-hop, jazz, metal, pop, reggae, and rock. The audio files are stored in the directory `genres_original`, with each genre represented as a separate folder containing `.wav` files. The dataset is collected to train a deep learning model for automatic music genre classification.

Load and Preprocess Data

Each audio file in the dataset is processed into Mel spectrograms, which serve as a visual representation of audio frequencies. The process begins with loading each `.wav` file using the `librosa` library, ensuring that the audio data is correctly read for further processing. Since music tracks vary in length, each audio file is divided into smaller overlapping chunks, with each segment lasting 4 seconds and overlapping by 2 seconds. This method increases the number of training samples and ensures that significant portions of the audio are not lost, allowing the model to learn more robust patterns.

Once the audio chunks are generated, Mel spectrograms are computed for each segment using `librosa.feature.melspectrogram`. These spectrograms capture essential frequency-based features that help differentiate music genres.

After generating spectrograms, labels corresponding to each chunk are assigned based on the genre of the original audio file. To prepare the dataset for training and evaluation, the entire dataset is split into training and testing sets in an 80:20 ratio. The training set is used to teach the model, while the testing set evaluates its generalization performance on unseen data.

Building Model

A Convolutional Neural Network (CNN) is implemented for the classification of music genres, leveraging its ability to extract spatial patterns from images, specifically Mel spectrograms derived from audio signals. The model is designed to capture both frequency and time-based features, which are crucial for distinguishing between different genres. The architecture begins with convolutional layers, which apply multiple filters to detect patterns in the input Mel spectrograms. These filters learn to recognize relevant musical characteristics such as pitch, rhythm, and texture by detecting edges and transitions in the frequency-time representation.

To enhance feature extraction, multiple convolutional layers are stacked, each followed

by an activation function, specifically ReLU (Rectified Linear Unit), which introduces non-linearity into the model, allowing it to learn complex relationships within the data. These convolutional layers are interspersed with max pooling layers, which serve to reduce the dimensionality of the extracted features while preserving essential information. Max pooling selects the most significant values from a specific window, helping the model focus on dominant patterns while discarding less important variations.

To further mitigate overfitting, dropout layers are introduced at key points in the architecture. During training, these layers randomly deactivate a fraction of neurons, ensuring that the model does not become overly dependent. This technique improves the model's robustness and prevents it from memorizing training examples instead of learning generalizable features.

After feature extraction, the output from the convolutional and pooling layers is passed to a flatten layer, which converts the multi-dimensional feature maps into a one-dimensional array. This transformation enables the model to connect spatially extracted features with fully connected layers that follow. The fully connected layers act as decision-making units, where the model learns complex relationships between the extracted audio features.

To optimize learning, the model is compiled using the Adam optimizer, which adaptively adjusts the learning rate for efficient convergence. The learning rate is set to 0.0001 to balance speed and stability during training. The model is trained over 30 epochs with a batch size of 32, ensuring that it learns effectively from the dataset while maintaining computational efficiency.

Evaluating Model

Once the model is trained, its performance is assessed using both the training and testing datasets. The primary evaluation metrics include loss and accuracy, where the training loss, validation loss, and accuracy scores provide insight into how well the model has learned to classify music genres. A confusion matrix is generated to visually represent the model's predictions, showing the number of correct and incorrect classifications across different genres. This heatmap visualization helps in identifying misclassifications and understanding which genres the model struggles with the most. Additionally, a classification report is generated, which includes precision, recall, and F1-score for each genre. These metrics offer a deeper analysis of the model's predictive ability, highlighting how well it distinguishes between similar genres and where improvements might be needed.

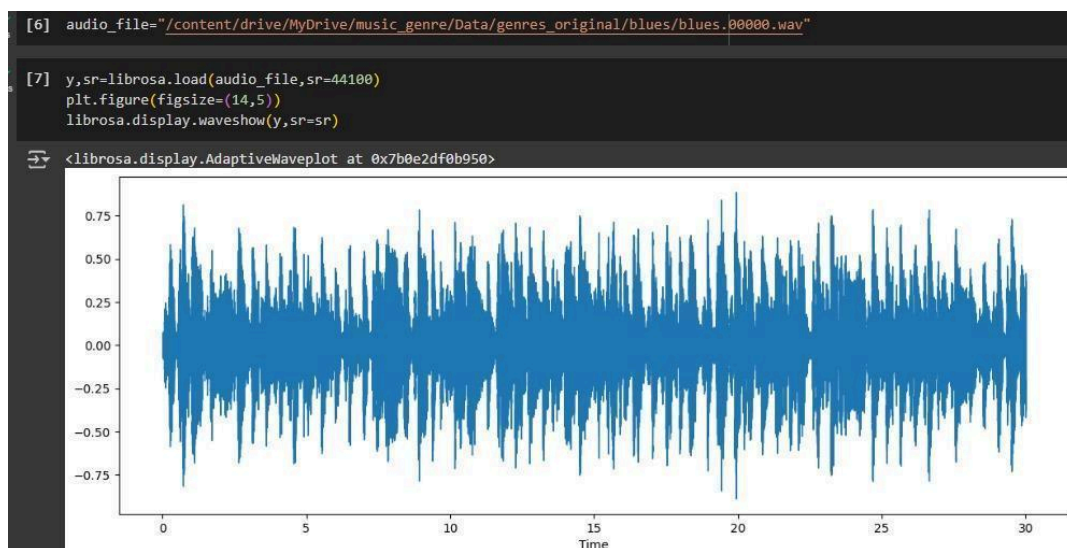
Testing the Model

To evaluate real-world performance, a dedicated function is implemented to predict the genre of an uploaded audio file in formats like .mp3 or .wav. The process begins by

loading the audio file and splitting it into smaller overlapping chunks to ensure sufficient data for analysis. For each chunk, a Mel spectrogram is computed to extract essential frequency-based features. These spectrograms are then passed through the trained model, which generates predictions for each chunk. The final genre classification is determined based on the most frequently predicted genre across all chunks, providing a reliable estimation of the audio's genre.

User-Interface

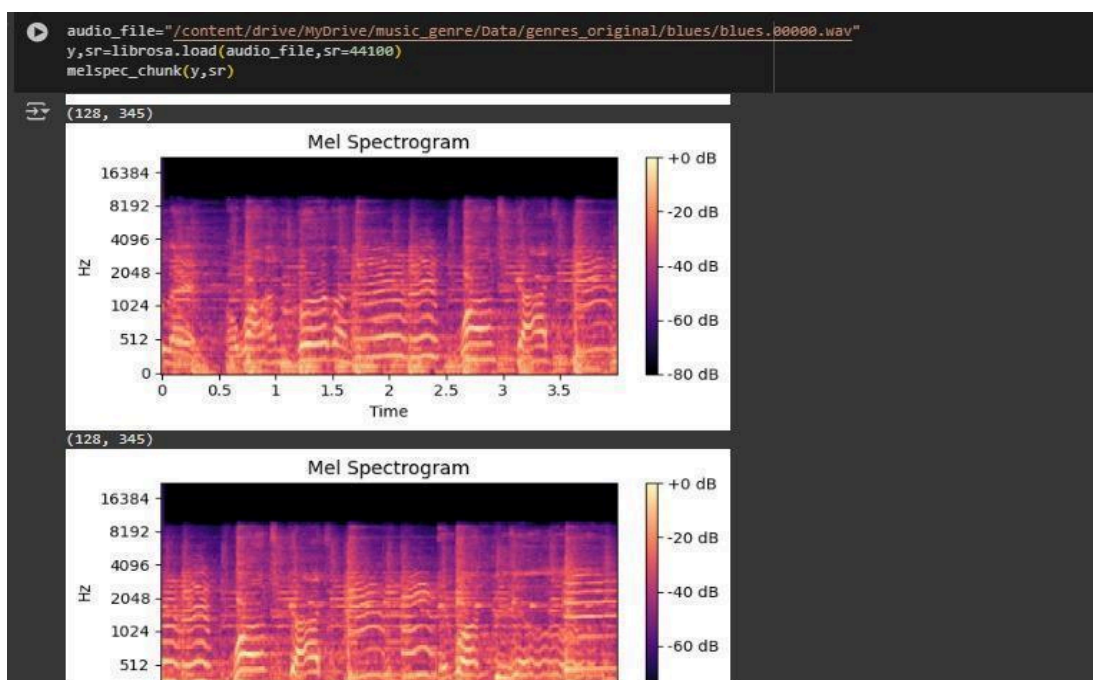
A Streamlit web application is developed to create an interactive interface for music genre classification. Users can upload an audio file in .mp3 or .wav format, which is then processed by the model to predict the genre. The application provides live predictions, displaying the identified genre along with a confidence score to indicate the model's certainty. A loading indicator with a spinner animation ensures a smooth user experience by visually signaling that the file is being processed.



5.2.1 Audio Track Visualization



5.2.2 Spectrogram Visualization



5.2.3 Chunks Spectrograms Visualization

```

load_and_preprocess_data(data_dir, classes, target_shape=(150, 150)):
data = []
labels = []
for i_class, class_name in enumerate(classes):
    class_dir = os.path.join(data_dir, class_name)
    print("Processing--", class_name)
    for filename in os.listdir(class_dir):
        if filename.endswith('.wav'):
            file_path = os.path.join(class_dir, filename)
            try:
                audio_data, sample_rate = librosa.load(file_path, sr=None)
            except Exception as e:
                print(f"Error processing {file_path}: {e}")
                continue
            # Preprocessing logic
            chunk_duration = 4
            overlap_duration = 2
            chunk_samples = chunk_duration * sample_rate
            overlap_samples = overlap_duration * sample_rate
            num_chunks = int(np.ceil((len(audio_data) - chunk_samples) / (chunk_samples - overlap_samples)))
            for i in range(num_chunks):
                start = i * (chunk_samples - overlap_samples)
                end = start + chunk_samples
                chunk = audio_data[start:end]
                mel_spectrogram = librosa.feature.melspectrogram(y=chunk, sr=sample_rate)

```

```

                mel_spectrogram = resize(np.expand_dims(mel_spectrogram, axis=-1), target_shape)
                data.append(mel_spectrogram)
                labels.append(i_class)

return np.array(data), np.array(labels)

```

5.2.4 Loading and Preprocessing


```
data, labels = data_preprocessing(data_dir, classes)
```

```
Processing-- blues  
Processing-- classical  
Processing-- country  
Processing-- disco  
Processing-- hiphop  
Processing-- jazz  
Processing-- metal  
Processing-- pop  
Processing-- reggae  
Processing-- rock
```

5.2.5 Audio Processing

```
X_train, X_test, y_train, y_test = train_test_split(data, labels, test_size=0.2, random_state=42)  
model = tf.keras.models.Sequential()  
print(X_train[0].shape)  
model.add(Conv2D(filters=32, kernel_size=3, padding='same', activation='relu', input_shape=X_train[0].shape))  
model.add(Conv2D(filters=32, kernel_size=3, activation='relu'))  
model.add(MaxPool2D(pool_size=2, strides=2))  
model.add(Conv2D(filters=64, kernel_size=3, padding='same', activation='relu'))  
model.add(Conv2D(filters=64, kernel_size=3, activation='relu'))  
model.add(MaxPool2D(pool_size=2, strides=2))  
model.add(Conv2D(filters=128, kernel_size=3, padding='same', activation='relu'))  
model.add(Conv2D(filters=128, kernel_size=3, activation='relu'))  
model.add(MaxPool2D(pool_size=2, strides=2))  
model.add(Dropout(0.3))  
model.add(Conv2D(filters=256, kernel_size=3, padding='same', activation='relu'))  
model.add(Conv2D(filters=256, kernel_size=3, activation='relu'))  
model.add(MaxPool2D(pool_size=2, strides=2))  
model.add(Conv2D(filters=512, kernel_size=3, padding='same', activation='relu'))  
model.add(Conv2D(filters=512, kernel_size=3, activation='relu'))  
model.add(MaxPool2D(pool_size=2, strides=2))  
model.add(Dropout(0.3))  
model.add(Flatten())  
model.add(Dense(units=1200, activation='relu'))  
model.add(Dropout(0.45))  
#Output layer  
model.add(Dense(units=len(classes), activation='softmax'))
```

5.2.6 Model Building

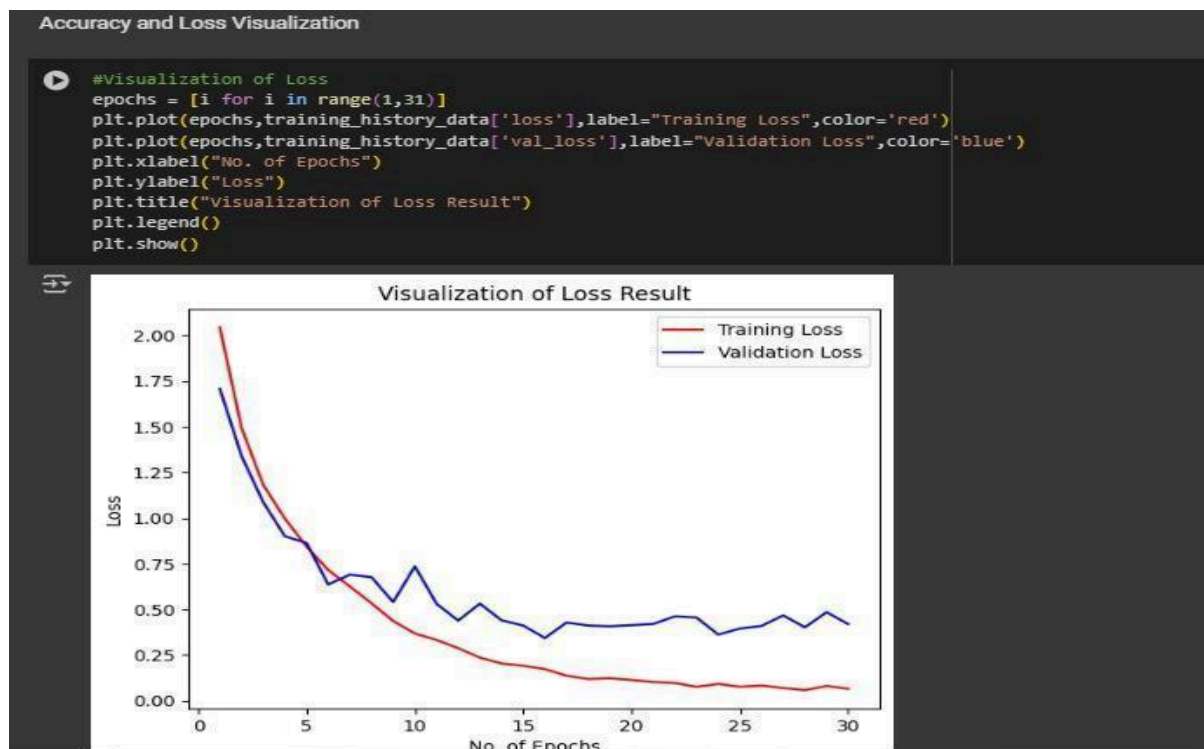
```
model.summary()
```

Model: "sequential"

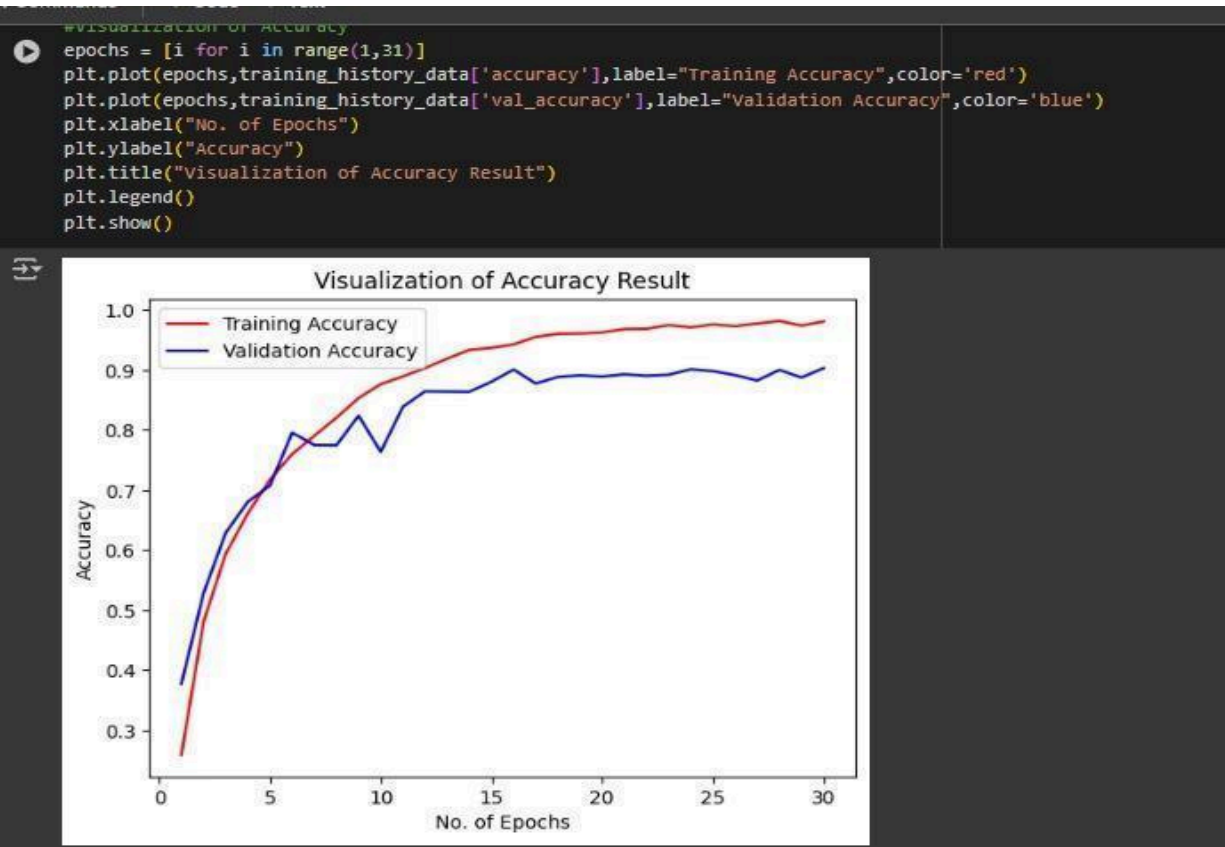
Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 150, 150, 32)	320
conv2d_1 (Conv2D)	(None, 148, 148, 32)	9,248
max_pooling2d (MaxPooling2D)	(None, 74, 74, 32)	0
conv2d_2 (Conv2D)	(None, 74, 74, 64)	18,496
conv2d_3 (Conv2D)	(None, 72, 72, 64)	36,928
max_pooling2d_1 (MaxPooling2D)	(None, 36, 36, 64)	0
conv2d_4 (Conv2D)	(None, 36, 36, 128)	73,856
conv2d_5 (Conv2D)	(None, 34, 34, 128)	147,584
max_pooling2d_2 (MaxPooling2D)	(None, 17, 17, 128)	0
dropout (Dropout)	(None, 17, 17, 128)	0
conv2d_6 (Conv2D)	(None, 17, 17, 256)	295,168
conv2d_7 (Conv2D)	(None, 15, 15, 256)	590,080
max_pooling2d_3 (MaxPooling2D)	(None, 7, 7, 256)	0
conv2d_8 (Conv2D)	(None, 7, 7, 512)	1,180,160
conv2d_9 (Conv2D)	(None, 5, 5, 512)	2,359,808
max_pooling2d_4 (MaxPooling2D)	(None, 2, 2, 512)	0
dropout_1 (Dropout)	(None, 2, 2, 512)	0
flatten (Flatten)	(None, 2048)	0
dense (Dense)	(None, 1200)	2,458,800
dropout_2 (Dropout)	(None, 1200)	0
dense_1 (Dense)	(None, 10)	12,010

Total params: 7,182,458 (27.40 MB)
Trainable params: 7,182,458 (27.40 MB)
Non-trainable params: 0 (0.00 B)

5.2.7 Model Summary



5.2.8 Loss result



5.2.9 Accuracy result

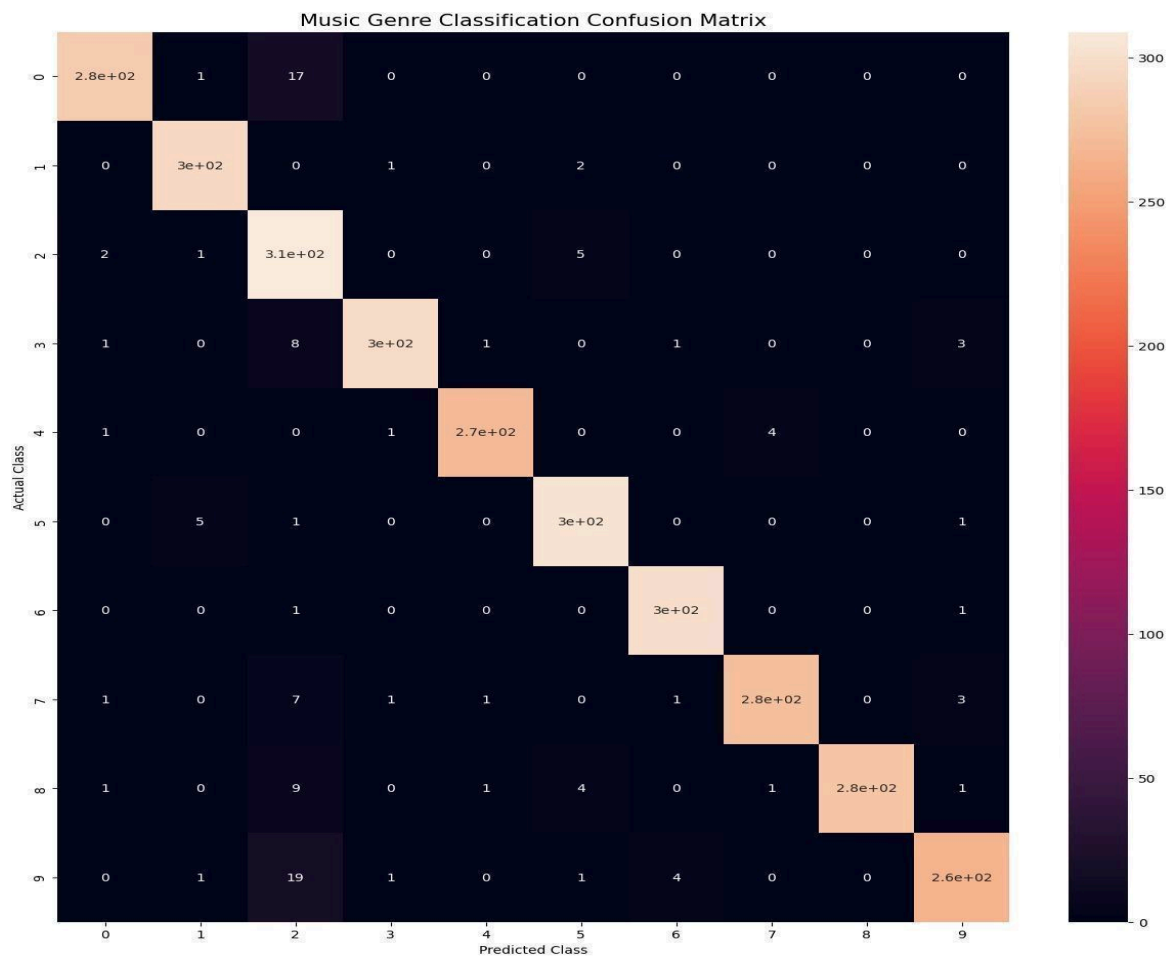
```

cm = confusion_matrix(true_categories,predicted_categories)
# Precision Recall F1score
print(classification_report(true_categories,predicted_categories,target_names=classes))

```

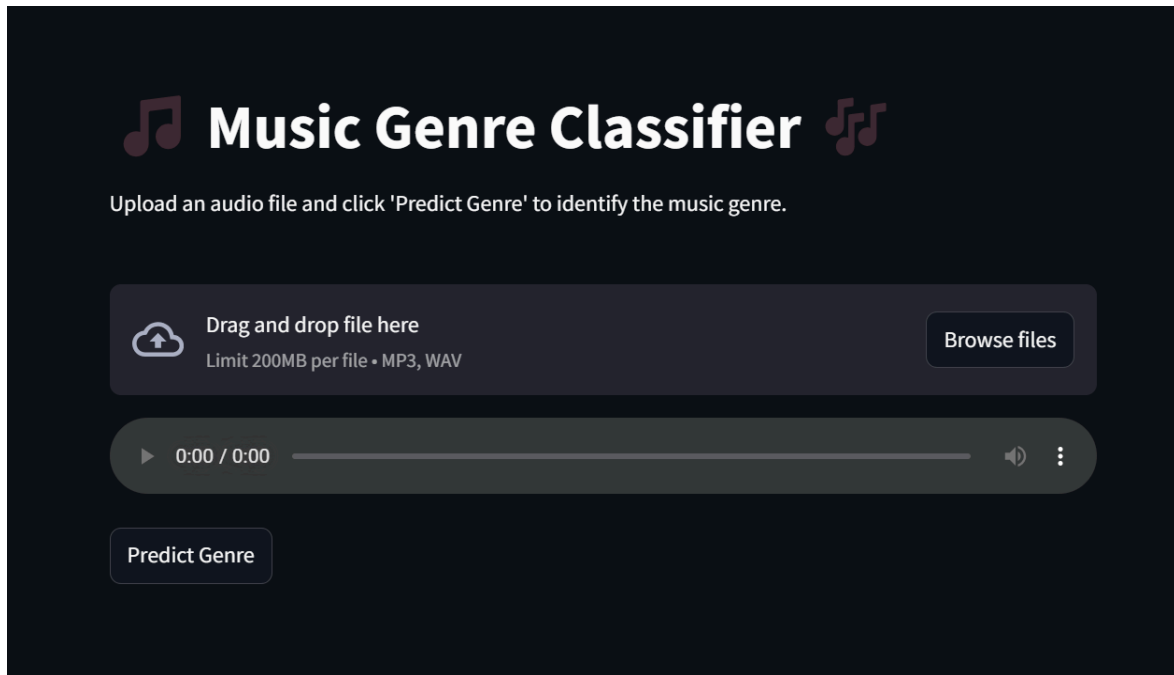
	precision	recall	f1-score	support
blues	0.98	0.94	0.96	302
classical	0.97	0.99	0.98	298
country	0.83	0.97	0.90	317
disco	0.99	0.96	0.97	312
hiphop	0.99	0.98	0.98	277
jazz	0.96	0.98	0.97	311
metal	0.98	0.99	0.99	302
pop	0.98	0.95	0.97	289
reggae	1.00	0.94	0.97	296
rock	0.97	0.91	0.94	291
accuracy			0.96	2995
macro avg	0.97	0.96	0.96	2995
weighted avg	0.96	0.96	0.96	2995

5.2.10 Classification Report

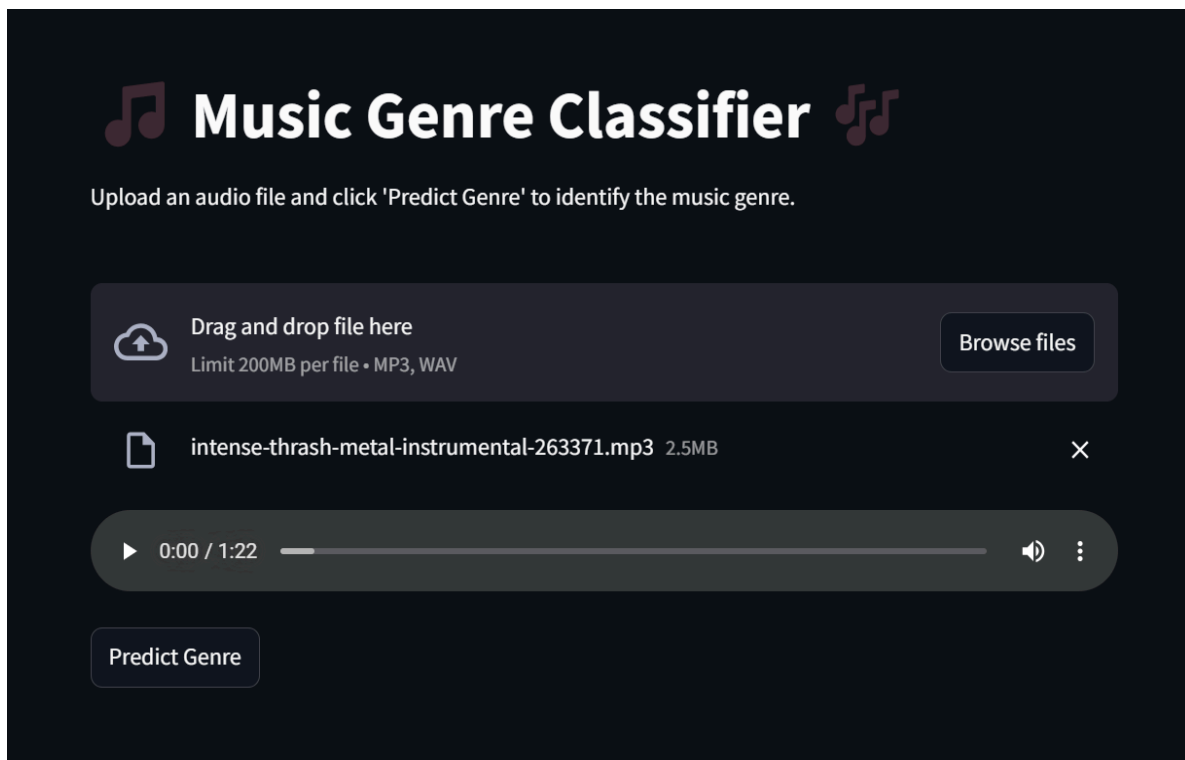


5.2.11 Confusion Matrix

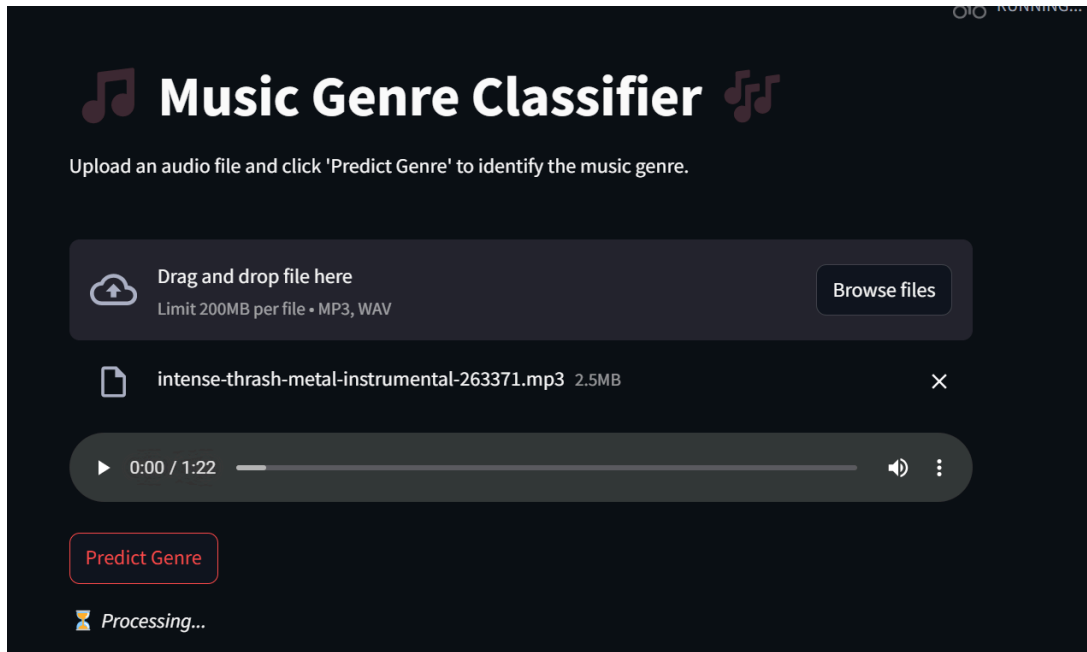
5.3 Interface Screens



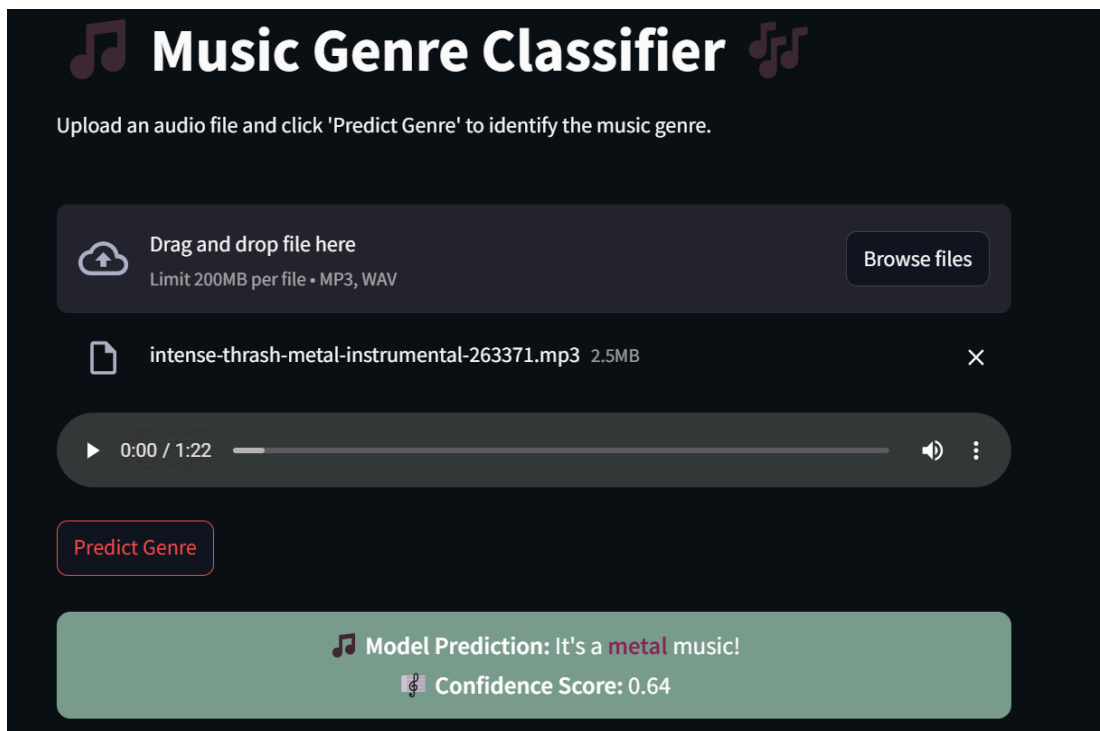
5.3.1



5.3.2



5.3.3



5.3.4

6. SYSTEM TESTING

6.1 Introduction

System testing is a vital phase in software and machine learning model development, ensuring that the entire system functions as expected and meets specified requirements. It is a black-box testing approach that evaluates the integrated system against functional and non-functional requirements before deployment. By simulating real-world scenarios, system testing helps identify defects, inconsistencies, and performance bottlenecks.

Testing involves two key processes: verification and validation. Verification ensures that the system is built correctly by checking compliance with design specifications, while validation ensures that the right system is built by confirming its alignment with user expectations. Both processes help in detecting errors early and improving overall system reliability.

For the Music Genre Classification for Predictive Analysis project, system testing ensures that the deep learning model accurately classifies audio files based on extracted features such as Mel spectrograms. The testing phase evaluates aspects such as data preprocessing, model inference, and real-time performance in a Streamlit-based deployment.

Machine learning models require a structured evaluation approach, typically involving a training set, validation set, and test set. The validation set helps tune hyperparameters to avoid overfitting, while the test set assesses the model's ability to generalize to new data. The test data should be diverse and representative of real-world scenarios to ensure the model performs reliably across different music genres.

System testing includes multiple methodologies, such as unit testing, integration testing, and performance testing, to examine different system components. Unit testing verifies the functionality of individual modules like feature extraction and model prediction, whereas integration testing ensures seamless interaction between components. Performance testing assesses the system's ability to handle large datasets and varied user inputs efficiently.

A well-structured testing process is essential to maintain the robustness of the system. By incorporating rigorous testing strategies, the Music Genre Classification for Predictive Analysis project ensures accurate predictions, optimized performance, and a seamless user experience. This phase plays a crucial role in validating the system's effectiveness and identifying areas for improvement before deployment.

Testing techniques encompass a variety of methods and approaches used to evaluate software or systems for defects and verify their correctness. Here's an overview of

different testing techniques commonly employed in software development:

- **Black Box Testing:**

This technique involves testing the functionality of a software application without examining its internal structure or code. Test cases are derived from the system's specifications and requirements, focusing on input-output behavior and ensuring that the software meets its functional requirements.
- **White Box Testing:**

Also known as structural testing or glass box testing, this technique examines the internal structure and code of the software being tested. Test cases are designed based on an understanding of the system's implementation, aiming to uncover defects in code logic, paths, and conditions.
- **Gray Box Testing:**

Combining elements of both black box and white box testing, gray box testing involves having partial knowledge of the internal workings of the software. Test cases are developed based on a combination of system specifications and an understanding of the code structure, allowing testers to target specific areas for testing.
- **Unit Testing:**

Unit testing focuses on testing individual units or components of the software in isolation. It verifies that each unit performs as expected and meets its specifications, helping to identify defects early in the development process.
- **Integration Testing;**

Integration testing evaluates the interactions between different units or components of the software. It verifies that integrated units function correctly together and that interfaces between components work as intended.
- **System Testing:**

System testing evaluates the behavior of the entire software system as a whole. It verifies that the system meets its functional and non-functional requirements, including performance, reliability, and usability, in a real-world environment.
- **Acceptance Testing:**

Acceptance testing involves evaluating the software's compliance with

business requirements and user expectations. It is typically performed by end-users or stakeholders to determine whether the software is ready for deployment.

- Regression Testing:

Regression testing verifies that changes or updates to the software do not introduce new defects or break existing functionality. It involves re-running previously executed test cases to ensure that no regression issues arise.

- Performance Testing:

Performance testing evaluates the software's responsiveness, scalability, and stability under various workload conditions. It assesses factors such as response times, throughput, and resource utilization to identify performance bottlenecks and optimize system performance.

- Security Testing:

Security testing assesses the software's resilience to security threats and vulnerabilities. It identifies potential security weaknesses such as unauthorized access, data breaches, or denial-of-service attacks and helps implement measures to mitigate risks.

These testing techniques are essential components of a comprehensive testing strategy, helping to ensure the quality, reliability, and security of software systems throughout the development lifecycle. By employing a combination of testing approaches, organizations can detect defects early, reduce development costs, and deliver high-quality software products to end-users.

Accuracy:

Accuracy is a key performance metric in evaluating the effectiveness of the Music Genre Classification for Predictive Analysis model. It measures the proportion of correctly predicted music genres out of the total predictions made by the system. A high accuracy score indicates that the deep learning model is well-trained and capable of distinguishing between different music genres based on extracted features.

In machine learning, accuracy is calculated as:

$$\text{Accuracy} = (\text{True Positives} + \text{True Negatives}) / (\text{Total Predictions})$$

where:

- **TP (True Positive):** Correctly predicted instances of a genre.
- **TN (True Negative):** Correctly predicted non-instances of a genre.
- **FP (False Positive):** Incorrectly classified instances.
- **FN (False Negative):** Missed correct classifications.

While accuracy is a useful metric, it may not always be the best indicator of model performance, especially when dealing with imbalanced datasets. To gain a deeper understanding of the model's effectiveness, additional evaluation metrics such as precision, recall, F1-score, and confusion matrix are analyzed.

Factors Affecting Accuracy

Several factors influence the accuracy of the genre classification model:

1. Feature Extraction Techniques:

The use of Mel spectrograms, MFCCs, and other audio features directly impacts the model's ability to differentiate genres.

2. Data Quality & Balance:

A diverse and well-balanced dataset ensures that the model does not favor one genre over others.

3. Model Selection:

Pre-trained models like CNNs and RNNs improve classification accuracy by capturing complex patterns in audio data.

4. Hyperparameter Tuning:

Optimizing parameters such as learning rate, batch size, and activation

functions can enhance accuracy.

5. Data Augmentation:

Techniques like time stretching, pitch shifting, and noise addition improve model generalization and prevent overfitting.

Evaluating Accuracy

The system's accuracy is evaluated using:

- **Train-Test Split:** The dataset is divided into training and testing sets, with the model trained on one set and evaluated on the other.
- **Cross-Validation:** K-fold cross-validation ensures robust accuracy estimation by averaging performance across multiple subsets.
- **Confusion Matrix:** A visual representation of correct and incorrect classifications for each genre.

Improving Accuracy

To enhance accuracy, the following strategies are implemented:

- **Increasing Data Diversity:** Expanding the dataset to include more music samples from various sources.
- **Fine-Tuning Pre-Trained Models:** Adjusting deep learning models like CNNs or transfer learning techniques for better performance.
- **Reducing Overfitting:** Using dropout layers, regularization, and early stopping to prevent the model from memorizing training data instead of learning patterns.

By continuously refining the model, optimizing parameters, and leveraging advanced deep learning techniques, the Music Genre Classification for Predictive Analysis project ensures high accuracy, making it a reliable tool for music classification and recommendation systems.

1. Test Plan(Planning Phase):

The test plan outlines the strategy for evaluating the Music Genre Classification for Predictive Analysis system to ensure it functions as expected. It defines the scope, objectives, environment, and criteria for successful testing. The primary goal is to verify that the system accurately classifies music genres based on extracted audio features. The testing scope includes validating the feature extraction process, genre classification accuracy, user interface responsiveness, and overall system performance. The test environment consists of a system with Python, TensorFlow/Keras, Librosa for audio processing, and Streamlite for deployment. The dataset used includes standard music

datasets such as GTZAN and FMA, along with real-world samples. Success is determined based on the system's ability to achieve a high accuracy rate, provide real-time classification, and maintain a seamless user experience.

2. Test Case Design(Preparation Phase):

Test case design involves creating specific test cases to assess different functionalities of the system. Each test case includes input data, expected output, actual results, and a pass/fail status. Important test cases include verifying file upload functionality, ensuring the system supports valid formats like .mp3 and .wav, and rejecting unsupported formats. Feature extraction tests ensure that the system correctly converts audio files into Mel spectrograms. Genre classification tests evaluate the model's ability to accurately predict genres across different audio samples. Additionally, user interface tests check the responsiveness and usability of the Streamlit application, while performance tests measure system response time under varying loads. These test cases help identify any inconsistencies or errors in the system.

3. Test Case Execution(Implementation Phase):

Test case execution involves running the designed test cases systematically and documenting the results. The process begins with uploading audio files to check file compatibility and handling. The feature extraction mechanism is then tested to confirm the proper generation of Mel spectrograms. The genre classification module is assessed by comparing predicted genres with actual ones to measure accuracy. Performance testing is conducted by evaluating response times when processing large files and noisy inputs. Throughout the execution, test results are recorded, and any failed test cases are analyzed for debugging and improvement. By thoroughly executing test cases, the system's reliability, accuracy, and efficiency are ensured before deployment.

6.2 Testing methods

System testing is a critical phase in software development where a fully integrated system is evaluated to ensure that all components function correctly as a unified whole. It verifies that the system meets specified requirements and performs reliably under real-world conditions. A software system is developed in individual modules, which are then integrated with other software components and hardware to form a complete system. System testing involves a comprehensive series of tests designed to validate the software's behavior, performance, and functionality from an end-user perspective.

The primary goal of system testing is to evaluate the end-to-end functionality of the application in an environment that closely resembles the production setup. This process involves navigating through all necessary modules to confirm that the system operates as expected. It ensures that the software works in conjunction with hardware components, third-party integrations, databases, and external interfaces.

There are two main approaches to software testing:

1. White Box Testing – This method focuses on the internal structure and logic of the software. Test cases are designed based on code implementation, and testers examine individual components, loops, conditions, and data flow to ensure correct execution.
2. Black Box Testing – This method tests the system from the user's perspective without examining the internal code structure. It involves testing inputs and outputs, functionality, user interface (UI), and system behavior under various conditions.

Unit Testing :

Unit testing focuses verification efforts on the smallest unit of software design—the module. It exercises specific paths in a module's control structure to ensure complete coverage and maximum error detection. This test is conducted on each module individually to ensure that it functions properly as an independent unit, which is why it is called Unit Testing.

During this testing phase, each module is examined separately, and the module interfaces are verified for consistency with design specifications. All important processing paths are tested to ensure the expected results. Unit testing is conducted during the development (coding phase) of an application, primarily by developers. It isolates sections of code and verifies their correctness. A unit may be an individual function, method, procedure, module, or object.

In the Software Development Life Cycle (SDLC) and Software Testing Life Cycle (STLC), unit testing is the first level of testing performed before integration testing. Unit testing

is a White Box testing technique usually performed by developers, but QA engineers may also conduct unit testing due to time constraints or developer reluctance.

Benefits of Unit Testing:

- Helps detect and fix bugs early in the development cycle, reducing costs.
- Assists developers in understanding the codebase and making changes efficiently.
- Provides valuable documentation for the project.
- Enhances code reuse—both the code and tests can be migrated to new projects with minimal changes.

Types of Unit Testing:

- **Manual Unit Testing** – Performed manually by developers or testers.
- **Automated Unit Testing** – Conducted using testing frameworks such as JUnit (Java), pytest (Python), NUnit (C#), and Mocha (JavaScript).

While unit testing can be done manually, automation is preferred in software engineering to ensure consistency and efficiency.

Integration Testing :

Integration testing evaluates how different modules interact with each other. Since software systems consist of multiple components working together, this type of testing ensures that these modules integrate and function correctly as a whole.

During this testing phase, data flow, API calls, and interactions between services are tested to detect potential integration issues, such as:

- Incorrect data exchange between modules.
- Broken communication due to API failures.
- Mismatched formats in data transmission.

Approaches to Integration Testing:

There are four types of integration testing approaches. Those approaches are the following:

- Big-Bang Integration Testing
- Bottom-Up Integration Testing
- Top-Down Integration Testing
- Mixed Integration Testing

Big-Bang Integration Testing:

It is the simplest integration testing approach, where all the modules are combining and verifying the functionality after the completion of individual module testing. In simple words, all the modules of the system are simply put together and tested. This approach is practicable only for very small systems. If once an error is found during the integration testing, it is very difficult to localize the error as the error may potentially belong to any of the modules being integrated. So, debugging errors reported during big bang integration testing is very expensive to fix. The Big-bang integration workflow diagram is a process diagram that shows how different parts of a system are integrated. It is typically used to show how different software components are integrated. It is a graphical representation of the software development process that is used in many organizations. It is a process that starts with the idea for a new software project and ends with the delivery of the software to the customer.

Bottom-Up Integration Testing:

Bottom-up Testing is a type of incremental integration testing approach in which testing is done by integrating or joining two or more modules by moving upward from bottom to top through control flow of architecture structure. In these, low-level modules are tested first, and then high-level modules are tested. This type of testing or approach is also known as inductive reasoning and is used as a synthesis synonym in many case. Bottom-up testing is user-friendly testing and results in an increase in overall software development. This testing results in high success rates with long-lasting results.

Processing :

Following are the steps that are needed to be followed during the processing :

1. Clusters are formed by merging or combining low-level modules or elements. These clusters are also known as builds that are responsible for performing the certain secondary or subsidiary function of a software.
2. It is important to write a control program for testing. These control programs are also known as drivers or high-level modules. It simply coordinates input and output of a test case.
3. Testing is done of the entire build or cluster containing low-level modules.
4. Lastly, control programs or drivers or high level modules are removed and clusters are integrated by moving upward from bottom to top in program structure with help of control flow.

Top-Down Integration Testing:

Top-down testing is a type of incremental integration testing approach in which testing is done by integrating or joining two or more modules by moving down from top to bottom through control flow of architecture structure. In these, high-level modules are tested first, and then low-level modules are tested. Then, finally, integration is done to ensure that the system is working properly. Stubs and drivers are used to carry out this project. This technique is used to increase or stimulate behavior of Modules that are not integrated into a lower level

Processing :

Following are the steps that are needed to be followed during processing :

1. Test driver represents the main control module also known as a high-level module and stubs are generally used for all low-level modules that directly subordinate (present or rank below another) to high-level modules.
2. In this, testing takes place from the bottom. So, high-level modules are tested first in isolation.
3. After this, low-level modules or subordinated modules or stubs replace high-level modules one by one at a time. This can be done using methods like depth-first breadth search.
4. The process is repeated until each module is integrated and tested.
5. Another stub replaces the present real or control module after completion of each set of tests. These stubs act as a temporary replacement for a called module (stubs) and give the same result or output as the actual product gives.
6. To check if there is any defect or any error occurring or present, regression testing is done

and it's important to reduce any side effects that might be caused due to errors occurring.

Validation Testing

Validation Testing ensures that the final product meets business requirements and user expectations. This phase checks whether the software behaves as expected under real-world conditions. It confirms that the software aligns with functional, non-functional, and business requirements. This type of testing is conducted after integration testing to verify that the complete system satisfies its intended use.

Validation testing ensures that:

- The software product meets customer expectations and specifications.

- The system is usable, efficient, and reliable in real-world conditions.
- The software works as expected under different environments and workloads.

Types of Validation Testing:

1. Functional Validation

Functional validation ensures that all required features and functionalities work correctly according to the software requirements. It validates that the software performs expected operations without errors or failures.

Key Aspects of Functional Validation:

- Testing each function by providing appropriate inputs and verifying the outputs.
- Ensuring that the software meets functional requirements specified in the documentation.
- Checking for correct data processing, calculations, and business logic execution.

Techniques Used in Functional Validation:

- Requirement-based testing – Ensures all specified functionalities are covered.
- Boundary value analysis – Tests edge cases to verify function stability.
- Equivalence partitioning – Divides input data into valid and invalid groups for testing.

2. System Validation

System Validation ensures that the entire system functions properly in an integrated environment. It verifies that software, hardware, and network components work together as intended.

Key Aspects of System Validation:

- Testing the overall performance, reliability, and stability of the system.
- Ensuring data integrity, security, and proper response to user interactions.
- Validating system behavior under different network conditions and environments.

Types of System Validation:

- **Load Testing** – Tests how the system handles a high volume of users.
- **Stress Testing** – Evaluates the system's performance under extreme conditions.
- **Security Testing** – Ensures that the system is protected from vulnerabilities.

3. User Acceptance Testing (UAT)

User Acceptance Testing (UAT) is the final phase of validation testing and is conducted by end-users to verify that the software meets their needs before deployment.

Key Aspects of UAT:

- Ensuring that the software meets business and user requirements.
- Testing under real-world scenarios with actual end-users.
- Gathering user feedback to determine if the system is ready for release.

Types of UAT:

- **Alpha Testing** – Conducted by internal testers before the product is released to real users.
- **Beta Testing** – Performed by a limited group of external users to gather feedback before the final release.
- **Regulation Acceptance Testing** – Ensures compliance with legal and industry standards.

Benefits of Validation Testing:

- Ensures the software meets customer expectations before deployment.
- Identifies defects that might have been overlooked in earlier testing phases.
- Reduces post-deployment failures and minimizes maintenance costs.
- Improves user satisfaction by ensuring a bug-free and functional product.
- Ensures that the software complies with business, functional, and legal requirements.

White Box Testing

White Box Testing (also called Glass Box or Structural Testing) focuses on analyzing the internal logic, code structure, and data flow of the application. It is a detailed testing method that evaluates code paths, conditions, loops, and statements to ensure that all possible execution scenarios are tested.

Techniques Used in White Box Testing:

- **Control Flow Testing** – Analyzes the sequence of execution in code.
- **Data Flow Testing** – Examines how data moves through the software.
- **Path Testing** – Identifies all possible paths through the program.

Benefits of White Box Testing:

- Helps optimize performance by identifying inefficient code.
- Detects security vulnerabilities within the software.
- Ensures full test coverage of the codebase.

Black Box Testing

Black Box Testing is a functional testing approach that evaluates software without examining its internal code. The focus is on inputs, outputs, and user interactions, ensuring that the application behaves correctly under different scenarios. Black box testing is commonly performed by QA testers and end-users to verify the software's functionality before release.

Techniques Used in Black Box Testing:

- **Boundary Value Analysis (BVA)** – Tests values at the edges of input ranges.
- **Equivalence Partitioning** – Divides inputs into valid and invalid groups to test representative cases.
- **Exploratory Testing** – Involves manual, unscripted testing to uncover hidden issues.

Benefits of Black Box Testing:

- Does not require knowledge of the internal code.
- Ensures that the system meets user requirements.
- Identifies usability and functionality issues from a real-world perspective.
- Helps uncover unexpected behavior when testing real-world scenarios.
- Provides a user-centric testing approach, improving software usability.
- Reduces the risk of bias since testers do not need to understand the code implementation.

6.3 Test Cases

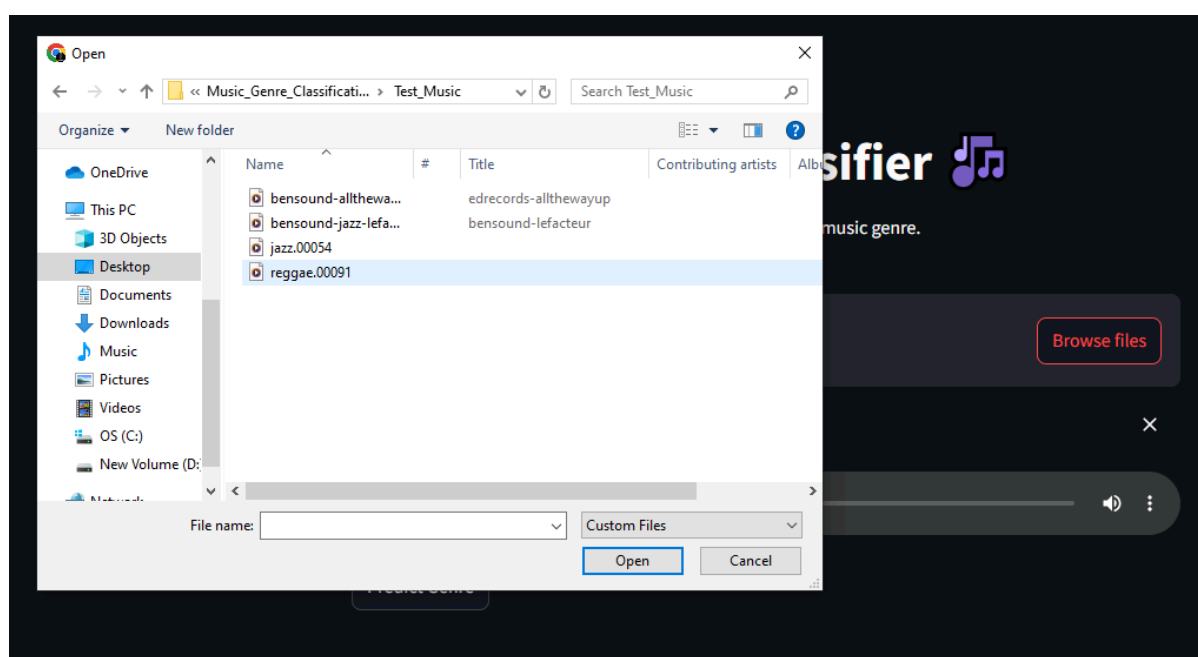


Fig 6.3.1 : Uploading Audio File

1. Uploading a Valid Audio File:

Test Scenario: The user uploads a valid audio file (MP3 or WAV format).

Test Steps:

- Open the Streamlit web interface.
- Click on the "Upload Audio File" button.
- Select and upload a valid music file.

Expected Output: The file should be successfully uploaded and displayed on the interface.

Actual Output: The file is uploaded correctly, and the system processes the audio

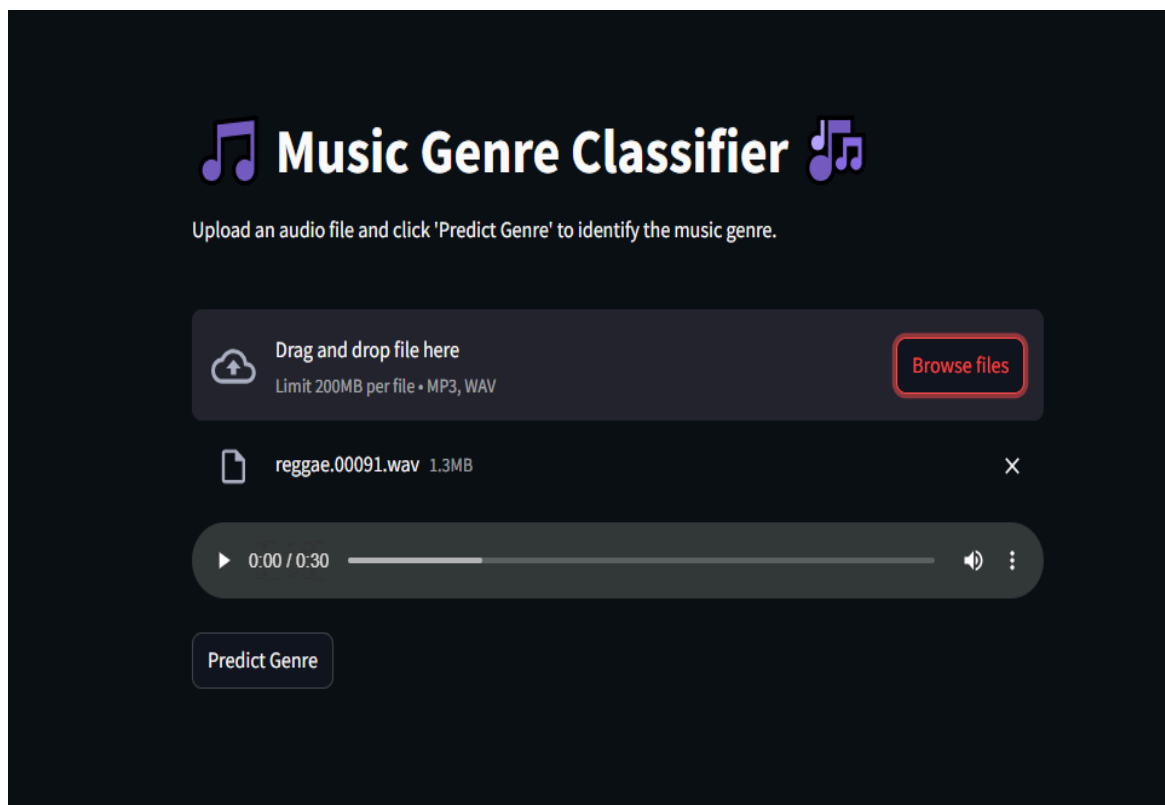


Fig 6.3.2: File Loaded successfully

2. Extracting Features from Audio File

Test Scenario: The system extracts Mel spectrogram features from the uploaded audio.

Test Steps:

- Upload a valid audio file.
- The system processes the audio and converts it into a spectrogram.
- The extracted features are displayed or stored for classification.

Expected Output: The Mel spectrogram should be generated without errors.

Actual Output: The spectrogram is successfully extracted, and feature values are stored.

```
2025-03-09 08:06:21.453 `label` got an empty value. This is discouraged f
n exception. Please provide a non-empty label and hide it with label_visi
Predicting Genre
1/1 ————— 1s 525ms/step
Audio is loaded successfully
Mel spectrogram generated
Mel spectrogram fed for cnn for training
```

Fig 6.3.3: Mel spectrogram Generated

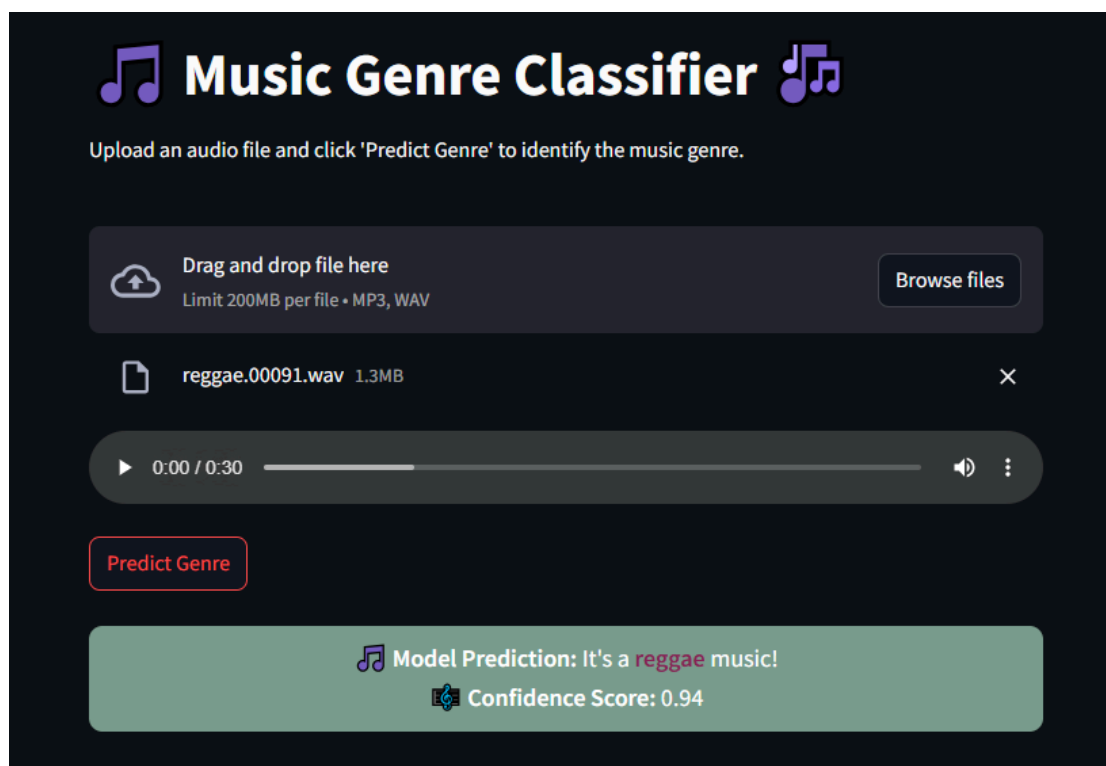


Fig 6.3.4: Predicted Genre

3. Classifying the Music Genre

Test Scenario: The system predicts the correct genre based on extracted audio features.

Test Steps:

- Upload an audio file and allow the system to process it.
- Click the "Predict Genre" button.
- The system analyzes the spectrogram and classifies the genre.

Expected Output: The correct genre label should be displayed on the interface.

Actual Output: The predicted genre is displayed, matching the expected genre with high accuracy.

7. CONCLUSION

The Music Genre Classification for Predictive Analysis project successfully demonstrates the effectiveness of deep learning techniques in identifying musical genres from audio files. By leveraging Mel spectrograms as input features, the model efficiently captures the unique frequency characteristics of different genres. The use of a Convolutional Neural Network (CNN) allows for deep feature extraction, enabling high-accuracy classification. The structured preprocessing approach, including chunking and feature scaling, further enhances the model's ability to generalize across various music samples.

Through rigorous training and evaluation, the model achieves reliable classification performance. Techniques such as dropout regularization and max pooling help prevent overfitting and improve generalization. The model's accuracy is validated using multiple performance metrics, including confusion matrices and classification reports.

The deployment of the model using Streamlit transforms it into a user-friendly application, allowing real-time genre classification. The web interface enhances accessibility, enabling users to upload audio files and receive instant results. The integration of visualization tools further helps in understanding how the classification is performed. Real-world testing on unseen audio tracks confirms the system's robustness, demonstrating its capability to handle diverse music styles effectively.

Future improvements to the project could involve expanding the dataset to include more diverse genres and incorporating advanced deep learning architectures for improved classification accuracy. Additional optimizations, such as reducing inference time and enhancing feature extraction techniques, would further refine the model's performance. By continuously improving and scaling the system, this project lays the foundation for more sophisticated applications in music analysis, recommendation systems.

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9. APPENDIX

APPENDIX-A

Project Repository Details

Project Title: Music Genre Classification for Predictive Analysis Using Deep Learning

Batch: A1

Batch Members:

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Submission Date: 09/04/2025

Project Repository Link The complete project files, including source code, documentation, and additional resources, are available at the following GitHub repository:

GitHub Repository:

https://github.com/atluridhatri/music_genre_classification_for_predictive_analysis_using_deep_learning.git

For quick access, scan the QR code below:



APPENDIX-B

GTZAN – A Widely Used Dataset for Music Genre Classification:

The GTZAN dataset is one of the most widely used benchmark datasets for music genre classification. It contains 1,000 audio tracks, each 30 seconds long, distributed across 10 different genres:

- Blues
- Classical
- Country
- Disco
- Hip-hop
- Jazz
- Metal
- Pop
- Reggae
- Rock

Each genre consists of 100 tracks, making it a balanced dataset. The dataset provides a diverse range of musical styles and has been extensively used in research on audio classification, deep learning, and feature extraction. However, it has certain limitations, such as the presence of duplicate recordings and some potential misclassifications, which researchers often address through additional data preprocessing and augmentation techniques.

The GTZAN dataset is one of the most popular and widely used datasets in the field of music genre classification. It was introduced in 2002 and has since become a benchmark for evaluating machine learning and deep learning models in this domain. The dataset consists of 1,000 audio tracks, each 30 seconds long, evenly distributed across 10 music genres: blues, classical, country, disco, hip-hop, jazz, metal, pop, reggae, and rock. Each genre contains 100 tracks, making the dataset balanced and suitable for training and testing classification models.

The audio files in GTZAN are in .wav format, which is ideal for processing and feature extraction. Researchers and developers often use this dataset to extract features such as Mel-frequency cepstral coefficients (MFCCs), chroma features, spectral contrast, and Mel spectrograms. These features are then fed into machine learning or deep learning models to classify the genre of the audio. Despite its popularity, the GTZAN dataset has some limitations, such as mislabeled tracks and a relatively small size compared to

modern datasets. However, it remains a valuable resource for benchmarking and prototyping music genre classification systems.

DL – Deep Learning:

Deep Learning (DL) is a subset of machine learning that focuses on training artificial neural networks with multiple layers to learn complex patterns and representations from data. Unlike traditional machine learning algorithms, which often require manual feature engineering, deep learning models can automatically extract relevant features from raw data, making them highly effective for tasks such as image recognition, natural language processing, and audio analysis.

Deep learning models are inspired by the structure and function of the human brain, consisting of interconnected layers of neurons. These models learn by adjusting the weights of connections between neurons during training, using techniques such as backpropagation and gradient descent. The depth of the network, i.e., the number of layers, allows the model to capture hierarchical features, starting from low-level patterns (e.g., edges in an image or frequencies in audio) to high-level abstractions (e.g., objects in an image or musical genres).

One of the key advantages of deep learning is its ability to handle large-scale datasets and perform end-to-end learning. For example, in music genre classification, deep learning models can take raw audio waveforms or spectrograms as input and directly output the predicted genre, eliminating the need for manual feature extraction. Popular deep learning architectures include Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Transformers, each suited for different types of data and tasks.

Deep learning has revolutionized many fields, including computer vision, speech recognition, and natural language processing. Its applications in audio analysis, such as music genre classification, have shown remarkable success due to the ability of deep learning models to capture intricate patterns in audio data. However, deep learning also comes with challenges, such as the need for large amounts of labeled data, high computational resources, and careful tuning of hyperparameters to achieve optimal performance.

Types of Deep Learning Models:

1. **Convolutional Neural Networks (CNNs):** CNNs are widely used for processing grid-like data, such as images and spectrograms. They use convolutional layers to extract spatial features, making them ideal for tasks like image classification, object detection, and audio analysis.

2. **Recurrent Neural Networks (RNNs):** RNNs are designed for sequential data, such as time series, text, or audio. They have memory capabilities that allow them to capture temporal dependencies, making them suitable for tasks like speech recognition, language modeling, and music generation.
3. **Transformers:** Transformers are a newer architecture that has gained popularity for natural language processing tasks. They use self-attention mechanisms to capture long-range dependencies in data, making them highly effective for tasks like machine translation, text summarization, and even audio analysis.
4. **Generative Adversarial Networks (GANs):** GANs consist of two neural networks—a generator and a discriminator—that compete against each other. They are used for generating realistic data, such as images, audio, or even music.

Performance Metrics:

Performance metrics are essential tools for evaluating the effectiveness of a machine learning or deep learning model. They provide quantitative measures of how well the model is performing in terms of accuracy, precision, recall, and other criteria. In the context of music genre classification, performance metrics help assess the model's ability to correctly identify the genre of an audio file and provide insights into areas where the model may need improvement.

One of the most commonly used metrics is accuracy, which measures the proportion of correctly classified instances out of the total number of instances. While accuracy is a straightforward and intuitive metric, it can be misleading in cases where the dataset is imbalanced. For example, if one genre dominates the dataset, a model might achieve high accuracy by simply predicting that genre most of the time, even if it performs poorly on other genres. To address this limitation, additional metrics such as precision, recall, and F1-score are often used.

Precision measures the proportion of true positive predictions (correctly predicted instances of a genre) out of all positive predictions made by the model. It is particularly useful when the cost of false positives is high. For example, in a music recommendation system, recommending the wrong genre to a user might lead to a poor user experience. On the other hand, recall measures the proportion of true positives out of all actual positives in the dataset. It is important when the goal is to minimize false negatives, such as ensuring that no relevant genres are missed in a classification task. The F1-score is the harmonic mean of precision and recall, providing a balanced measure of the model's performance, especially in imbalanced datasets.

Another important metric is the confusion matrix, which provides a detailed breakdown of the model's predictions. It shows the number of correct and incorrect predictions for each genre, allowing for a more granular analysis of the model's strengths and weaknesses. For instance, the confusion matrix might reveal that the model struggles to distinguish between genres with similar acoustic characteristics, such as jazz and blues. This insight can guide further improvements, such as refining the feature extraction

process or augmenting the dataset with more examples of those genres.

In addition to these metrics, ROC-AUC (Receiver Operating Characteristic - Area Under the Curve) is often used to evaluate the model's ability to distinguish between classes. The ROC curve plots the true positive rate (recall) against the false positive rate at various threshold settings, and the AUC provides a single value summarizing the model's performance across all thresholds. A higher AUC indicates better performance, with a value of 1 representing a perfect classifier.

Finally, log loss (or cross-entropy loss) is a metric commonly used in classification tasks, especially when the model outputs probabilities for each class. It penalizes incorrect predictions more heavily if the model is confident in its wrong predictions, making it a useful metric for evaluating the calibration of the model's confidence. For example, if the model assigns a high probability to the wrong genre, the log loss will be significantly higher than if it had assigned a lower probability.

CNN – Convolutional Neural Network:

Convolutional Neural Networks (CNNs) are a class of deep learning models specifically designed for processing structured grid data, such as images and spectrograms. CNNs are widely used in computer vision tasks, but they have also proven to be highly effective for audio analysis, including music genre classification.

The key component of a CNN is the convolutional layer, which applies a set of filters (or kernels) to the input data to extract local features. For example, in image processing, these filters might detect edges, textures, or shapes. In audio analysis, CNNs are often applied to spectrograms, which are visual representations of audio signals. The filters in this case can capture patterns such as harmonic structures, rhythmic elements, or timbral characteristics that are indicative of specific music genres.

CNNs typically consist of multiple convolutional layers, followed by pooling layers that reduce the spatial dimensions of the feature maps, and fully connected layers that combine the extracted features to make predictions. The use of convolutional layers allows CNNs to be translation-invariant, meaning they can detect patterns regardless of their position in the input data. This property is particularly useful for audio analysis, where the same musical pattern might occur at different time points in a track.

In music genre classification, CNNs have become a popular choice due to their ability to automatically learn relevant features from spectrograms and achieve high accuracy. For instance, a CNN model might take a Mel spectrogram as input, pass it through several convolutional and pooling layers to extract features, and then use a softmax layer to predict the genre. The success of CNNs in this domain has led to their widespread adoption in both research and industry for audio-related tasks.

How CNNs Work

The core building block of a CNN is the convolutional layer. This layer applies a set of filters (also called kernels) to the input data to extract local features. Each filter slides over the input, performing element-wise multiplication and summation to produce a

feature map. These feature maps capture patterns such as edges, textures, or shapes in images, or frequency and temporal patterns in spectrograms. For example, in music genre classification, a CNN might use filters to detect harmonic structures, rhythmic patterns, or timbral characteristics that are unique to specific genres.

Following the convolutional layers, pooling layers are often used to reduce the spatial dimensions of the feature maps. Pooling operations, such as max pooling or average pooling, summarize the features in a local region, making the model more computationally efficient and robust to small variations in the input. For instance, max pooling retains the most prominent feature in a region, which helps the model focus on the most important patterns.

After several convolutional and pooling layers, the extracted features are passed through fully connected layers (also called dense layers). These layers combine the features to make predictions. In a classification task, the final layer typically uses a softmax activation function to output probabilities for each class.

The main advantage of CNN is that CNNs can learn relevant features directly from raw data, eliminating the need for manual feature engineering. This is especially useful in tasks like music genre classification, where the relevant features (e.g., rhythm, harmony, timbre) can be complex and domain-specific.

CNNs are capable of detecting patterns regardless of their position in the input. For example, a musical motif can be recognized whether it occurs at the beginning, middle, or end of a track.

CNNs learn features in a hierarchical manner. Early layers capture low-level features (e.g., edges or simple frequencies), while deeper layers capture high-level features (e.g., complex musical structures or genre-specific patterns).